

Microeconomic Theory I

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Fall 2026 edition

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Preface to the Fall 2026 Edition

These notes provide a rigorous first course in graduate microeconomic theory. The Fall 2026 edition preserves the substantive coverage of the Fall 2024 course while improving the exposition, notation, organization, and portability of the source files. The central discipline throughout is to distinguish primitives, behavioral assumptions, optimization problems, equilibrium conditions, and empirical implications.

The material is cumulative. Part I develops the mathematical language of binary relations and connects preferences to observable choice. Part II derives individual demand and extends the benchmark model to limited attention and uncertainty. Part III studies aggregation and stochastic choice. Part IV develops production, competitive markets, and general equilibrium. Part V treats equilibrium restrictions, trade under uncertainty, matching, and computable general equilibrium.

Notation and Conventions

The following conventions apply unless a chapter explicitly states otherwise. They are collected here so that the same symbol does not silently change meaning across topics.

- There are L commodities, indexed by $\ell = 1, \dots, L$. Commodity bundles are column vectors $x \in X \subseteq \mathbb{R}_+^L$.
- Consumers are indexed by $i = 1, \dots, I$; firms are indexed by $j = 1, \dots, J$. Consumption is denoted by x , production by y , and endowments by ω .
- Prices are column vectors $p \in \mathbb{R}_{++}^L$, unless a result explicitly permits zero boundary prices. Set difference is denoted by $A \setminus B$. The operators $\max$ and $\min$ are typeset as mathematical operators. The scalar $w > 0$ denotes wealth. The budget set is $B(p, w) = \{x \in X : p \cdot x \leq w\}$.
- Marshallian demand is $x(p, w)$, Hicksian demand is $h(p, \bar{u})$, indirect utility is $v(p, w)$, and the expenditure function is $e(p, \bar{u})$.
- The dot $\cdot$ denotes the Euclidean inner product, and $\top$ denotes transpose. A prime marks an alternative object, as in p' or x' ; it is not used for transpose.
- Weak preference, strict preference, and indifference are denoted by $\succeq$, $\succ$, and $\sim$, respectively. Choice rules are denoted by c or C only when the surrounding section defines the domain and whether the rule is single- or set-valued.
- For a differentiable vector-valued function, $D_p x$ is its Jacobian with respect to prices, while ∇u is the column gradient of a scalar-valued function.

Part I

Foundations of Choice and Demand

Chapter 1

Basics of Consumer Theory

1.1 Preliminaries

Consumer theory begins with feasible choice and observable demand. This chapter introduces the commodity space, budget sets, demand, Walras' law, homogeneity of degree zero, the weak axiom of revealed preference, and the Slutsky matrix. The next chapter develops the abstract connection between choice and preference; the utility-maximization chapter then joins these two perspectives.

1.2 Environment: Commodities and Budget Set

An economic commodity is a good or service whose quantity matters to the consumer. There are $L \geq 1$ commodities. A consumption bundle is a column vector $x = (x_1, \dots, x_L)^\top \in \mathbb{R}^L$, and x_ℓ denotes the quantity of commodity $\ell \in \{1, \dots, L\}$.

The consumption set $X \subseteq \mathbb{R}^L$ records the bundles that are feasible for the consumer. Unless a section states otherwise, we take $X = \mathbb{R}_+^L$, so negative consumption is ruled out. Other applications may impose upper bounds or interpret coordinates as net trades.

The consumer cannot generally choose every bundle in X . In the benchmark model, prices and wealth are the only constraints. Let $w > 0$ denote wealth and let $p \in \mathbb{R}_{++}^L$ denote the price vector. The budget set is $B(p, w) = \{x \in X : p \cdot x \leq w\}$, a subset of the consumption set.

1.3 Demand Functions

After specifying the consumer's environment, we take demand as the observable primitive. A demand function is a map $x : \mathbb{R}_{++}^L \times \mathbb{R}_{++} \rightarrow X$ such that $x(p, w) \in B(p, w)$ for every price--wealth pair; equivalently, $p \cdot x(p, w) \leq w$.

The coordinate $x_\ell(p, w)$ denotes the quantity of commodity ℓ consumed at prices and wealth (p, w) .

Demand describes behavior; it does not yet explain that behavior through preferences or optimization. Treating the entire function as observed is an idealization, because an empirical dataset contains only finitely many price--wealth observations. We therefore distinguish restrictions on a full demand function from tests that can be implemented on finite data.

Example 1.1. Cobb-Douglas Demand Function. $x(p, w) = (\frac{\alpha w}{p_1}, \frac{(1-\alpha)w}{p_2})^\top$ for $\alpha \in (0, 1)$.

1.4 Consistency in Consumption

With the environment and demand primitive in place, we turn to consistency in consumption choices. This approach restricts observable behavior before imposing a specific preference representation.

We will impose a set of restrictions or axioms over the Marshallian demand function that we expect to hold in all situations irrespective of the details. Here we understand modeling consumption as restricting the set of allowable behaviors.

These axioms are minimal theoretical requirements, not empirical laws that every dataset must satisfy. Their value is that they make the set of admissible behavior explicit and provide a baseline against which richer models can be compared.

Axiom 1.2. (*Walras' law*) A demand function x is said to satisfy Walras' law if and only if $p \cdot x(p, w) = w$ for all price-wealth pairs.

Walras' law imposes budget exhaustion; it rules out choices that leave unspent wealth when a locally nonsatiated consumer could purchase more.

Axiom 1.3. (*Homogeneity of degree zero, HD0.*) A demand function x is homogeneous of degree zero if, for every $\alpha > 0$, $x(\alpha p, \alpha w) = x(p, w)$.

HD0 says that multiplying all prices and wealth by the same positive scalar does not change demand. It is a normalization of units, not a behavioral claim about intelligence.

Axiom 1.4. (*Weak axiom of revealed preference, WARP.*) A demand function x satisfies WARP if, for any two price-wealth pairs (p, w) and (p^*, w^*) , whenever $x(p, w) \neq x(p^*, w^*)$ and $p \cdot x(p^*, w^*) \leq w$, then $p^* \cdot x(p, w) > w^*$.

WARP excludes a two-observation reversal: if the bundle chosen at the second budget was affordable at the first, then the distinct bundle chosen at the first must be unaffordable at the second.

For an observed choice (p^t, x^t) , define direct revealed preference by R^D , where $x^t R^D y$ means that the observed bundle cost at least as much as y at the prices under which x^t was chosen:

$$x^t R^D y \iff p^t \cdot x^t \geq p^t \cdot y.$$

For distinct chosen bundles z and y , WARP is equivalently $z R^D y \implies \neg(y R^D z)$. This is a restriction on observed choices; it need not be interpreted as a primitive psychological preference relation.

1.5 Comparative Statics and Slutsky Matrix

After establishing the axioms of behavior, the next step is to study their implications for the parameters of the model. The parameters of the model in our study of consumer behavior are prices and wealth pairs. Assume throughout this section that the demand function is continuously differentiable on an open domain of strictly positive prices and wealth.

- The price Jacobian is $D_p x(p, w) \in \mathbb{R}^{L \times L}$; its (ℓ, k) entry is $\partial_{p_k} x_\ell(p, w)$.
- We let $D_w x(p, w) = (\partial_w x_\ell(p, w))_{\ell=1}^L$ be the vector of wealth effects.
- The Slutsky matrix is $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$. It measures the derivative of demand under a price change that adjusts wealth just enough to keep the previously chosen bundle affordable. Its (ℓ, k) entry is $S_{\ell k} = \partial_{p_k} x_\ell(p, w) + \partial_w x_\ell(p, w) x_k$.

Fix a reference bundle q and consider the compensated demand $x(p, p \cdot q)$. The chain rule gives $D_p[x(p, p \cdot q)] = D_p x(p, p \cdot q) + D_w x(p, p \cdot q) q^\top$. At $q = x(p, w)$, Walras' law gives $p \cdot q = w$, so this derivative is exactly the Slutsky matrix: $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$.

Implications of Walras' Law for the Slutsky Matrix

Fact 1.5. If x satisfies Walras' law, then (i) $p^\top D_w x(p, w) = 1$, (ii) $p^\top D_p x(p, w) = -x(p, w)^\top$, (iii) $p^\top S(p, w) = 0$.

Part (i) implies that at least one good has a positive wealth effect. Because prices are strictly positive and $\sum_{\ell} p_{\ell} \partial_w x_{\ell}(p, w) = 1$, there must be some ℓ such that $\partial_w x_{\ell}(p, w) > 0$. In general, Walras' law restricts the wealth effects, while part (iii) restricts consumption changes under Slutsky-compensated price changes.

Implications of Homogeneity of Degree Zero.

Claim 1.6. If x satisfies Walras' law and homogeneity of degree zero, then $S(p, w)p = 0$.

Proof. For fixed (p, w) , homogeneity gives $x(\lambda p, \lambda w) = x(p, w)$ for all $\lambda > 0$. Differentiating with respect to λ at $\lambda = 1$ yields

$$D_p x(p, w)p + D_w x(p, w)w = 0.$$

Walras' law gives $w = p \cdot x(p, w) = x(p, w)^{\top} p$; therefore the left-hand side is $S(p, w)p$, as required. $\square$

Implications of WARP.

The main implication of the WARP while assuming Walras' law is the Compensated Law of Demand (CLD).

Definition 1.7. (CLD) We say that x satisfies the Compensated Law of Demand if and only if $(p^* - p) \cdot (x(p^*, w^*) - x(p, w)) < 0$ when $x(p^*, w^*) \neq x(p, w)$ and $w^* = p^* \cdot x(p, w)$.

We next establish that WARP implies the compensated law of demand.

Proposition 1.8. *Let x satisfy Walras' law and WARP. Then x satisfies the CLD.*

Proof. Fix two price--wealth pairs satisfying $w^* = p^* \cdot x(p, w)$ and $x(p^*, w^*) \neq x(p, w)$. Set $z = x(p, w)$ and $y = x(p^*, w^*)$.

Walras' law and the compensation condition give $p \cdot z = w$ and $p^* \cdot z = p^* \cdot y = w^*$.

Because $p^* \cdot z \leq w^*$, WARP and $y \neq z$ imply $p \cdot y > w$, hence $p \cdot (y - z) > 0$.

Combining this strict inequality with $p^* \cdot (y - z) = 0$ gives

$$(p^* - p) \cdot (y - z) < 0,$$

$\square$

The differential version yields the following corollary.

Corollary 1.9. *Suppose x is continuously differentiable and satisfies Walras' law and WARP. Then, for every $a \in \mathbb{R}^L$ and every price-wealth pair, $a^\top S(p, w)a \leq 0$. Equivalently, the symmetric part $[S(p, w) + S(p, w)^\top]/2$ is negative semidefinite.*

Proof. Fix a price-wealth pair (p, w) and a direction $a \in \mathbb{R}^L$. For sufficiently small t , define $p(t) = p + ta$ and compensate wealth by $w(t) = p(t) \cdot x(p, w)$.

$$ta \cdot [x(p(t), w(t)) - x(p, w)] \leq 0.$$

$$\frac{a \cdot [x(p(t), w(t)) - x(p, w)]}{t} \leq 0$$

for every nonzero sufficiently small t after dividing the revealed-preference inequality by $t^2 > 0$. Continuous differentiability and the chain rule imply

$$\lim_{t \rightarrow 0} \frac{x(p(t), w(t)) - x(p, w)}{t} = D_p x(p, w)a + D_w x(p, w) x(p, w)^\top a = S(p, w)a.$$

Taking limits gives $a^\top S(p, w)a \leq 0$.

□

Marshallian and compensated price effects

The Slutsky decomposition separates the total response of demand to a price change from the response at fixed purchasing power. For a differentiable demand function,

$$\partial_{p_k} x_\ell(p, w) = S_{\ell k}(p, w) - x_k(p, w) \partial_w x_\ell(p, w).$$

Under WARP and Walras' law the compensated matrix has a negative-semidefinite symmetric part. Thus a positive Marshallian own-price derivative $\partial_{p_\ell} x_\ell > 0$ is possible only when the income effect is sufficiently adverse: the good is inferior ($\partial_w x_\ell < 0$), and the income effect dominates the nonpositive compensated effect. This is the Giffen mechanism.

The possibility of a Giffen response does not by itself establish a utility representation. Two distinct rationalizability routes are developed later. For a differentiable demand system, local integrability uses Walras' law, homogeneity, Slutsky symmetry and negative semidefiniteness together with explicit domain and regularity hypotheses. For a finite budget dataset, observed expenditure exhaustion and GARP provide the Afriat certificate. The compensated response, rather than the Marshallian derivative alone, is the object restricted by revealed preference.

1.6 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 1.1 [Core]. Budget scaling and Cobb-Douglas demand.

Let $X = \mathbb{R}_+^2$ and $B(p, w) = \{x \in X : p \cdot x \leq w\}$, where $p \gg 0$ and $w > 0$.

(a) Prove that $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$. Explain why every demand correspondence generated by preference maximization is homogeneous of degree zero in (p, w) .

(b) Suppose preferences are locally nonsatiated. Prove that every chosen bundle satisfies Walras' law, $p \cdot x = w$.

(c) For $u(x) = x_1^\alpha x_2^{1-\alpha}$ with $0 < \alpha < 1$, derive Marshallian demand and verify both properties directly.

Problem 1.2 [Core]. A two-observation WARP test.

A consumer chooses $x^1 = (2, 2)$ at $(p^1, w^1) = ((2, 1), 6)$ and $x^2 = (1, 5/2)$ at $(p^2, w^2) = ((1, 2), 6)$.

(a) Verify that each observed bundle exhausts its budget.

(b) Determine whether each bundle was affordable at the other observation.

(c) Does the dataset satisfy WARP for single-valued demand? Identify the two direct revealed-preference comparisons that determine your answer.

Problem 1.3 [Proof]. Differential restrictions on demand.

Let $x : \mathbb{R}_{++}^L \times \mathbb{R}_{++} \rightarrow \mathbb{R}_+^L$ be continuously differentiable and satisfy Walras' law.

(a) Differentiate Walras' law with respect to wealth and prices.

(b) Define the Slutsky matrix by $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$. Prove that $p^\top S(p, w) = 0$.

(c) Explain which additional restrictions on S follow from utility maximization and which of them do not follow from Walras' law alone.

Chapter 2

Preference and Choice

2.1 From Choice Data to Preference

Choice data, preference relations, and utility representations are three descriptions of the same decision problem. This chapter fixes the logical relations among them before prices and wealth are introduced in the consumer-theory chapter.

2.2 Choice environments

Let X be a nonempty set of alternatives and let $\mathcal{A} \subseteq 2^X \setminus \{\emptyset\}$ be the family of feasible menus.

Definition 2.1. Choice correspondence. A choice correspondence is a map $c : \mathcal{A} \rightrightarrows X$ satisfying $\emptyset \neq c(A) \subseteq A$ for every $A \in \mathcal{A}$. A choice structure is the pair $(\mathcal{A}, c)$.

Rationalization. A binary relation $\succeq$ rationalizes $(\mathcal{A}, c)$ if $c(A) = \{x \in A : x \succeq y \text{ for every } y \in A\}$ for every $A \in \mathcal{A}$. It weakly rationalizes the data if the equality is replaced by inclusion. A rational preference relation is complete and transitive.

2.3 Preference relations

Definition 2.2. For a relation $\succeq \subseteq X \times X$, define its strict and symmetric parts by $x \succ y \iff x \succeq y$ and not $y \succeq x$ and $x \sim y \iff x \succeq y$ and $y \succeq x$. If $\succeq$ is complete and transitive, then $\succ$ is irreflexive and transitive, while $\sim$ is an equivalence relation. These elementary implications are the order-theoretic basis for utility representation.

Proposition 2.3. *Finite utility representation. If X is finite, a preference relation on X has a*

real-valued utility representation if and only if it is complete and transitive. A representation is unique up to a strictly increasing transformation of its range.

For an infinite commodity or choice space, representation requires additional assumptions such as continuity and a suitable countability condition. The continuous-utility theorem used later is stated in the consumer-theory chapter.

2.4 Revealed preference and WARP

Definition 2.4. Direct revealed preference. For a choice structure $(\mathcal{A}, c)$, write $x \succeq^D y$ if there is a menu $A \in \mathcal{A}$ with $x \in c(A)$ and $y \in A$. Write $x \succ^D y$ if, in addition, $y \notin c(A)$. Let $\succeq^*$ be the reflexive-transitive closure of $\succeq^D$.

Weak Axiom of Revealed Preference (WARP). The data satisfy WARP when $x \succeq^D y$ implies not $y \succ^D x$. Equivalently, if two menus contain both x and y , choosing x from one and strictly choosing y from the other is ruled out.

Proposition 2.5. *Necessity of WARP. If a complete and transitive preference relation rationalizes a choice structure, then the structure satisfies WARP.*

Proof. If $x \in c(A)$ and $y \in A$, rationalization gives $x \succeq y$. If a second menu strictly chose y over x , it would give $y \succ x$, contradicting the first comparison.

2.5 GARP on finite data

Definition 2.6. Generalized Axiom of Revealed Preference (GARP). A finite dataset satisfies GARP when $x \succeq^* y$ never occurs together with $y \succ^D x$. Thus an indirect revealed-preference chain may return to an alternative only when the return comparison is not strict.

Theorem 2.7. *Finite-data rationalizability. See [Afriat, 1967, Varian, 1982]. For a finite budget dataset with positive prices and observed expenditure exhaustion, GARP is equivalent to weak rationalizability by a complete and transitive preference relation on the observed budgets. Afriat's theorem strengthens this conclusion to a locally nonsatiated, continuous and concave utility representation. The proof is deferred to the Afriat section of the consumer-theory chapter, where the finite inequalities are written explicitly.*

WARP is the one-step restriction. GARP applies the no-strict-reversal requirement to the reflexive-transitive closure of direct revealed preference and is therefore the relevant condition

when the data do not contain every pair of alternatives in a common menu. Exact single-valued rationalization can require additional tie-breaking assumptions, so the theorem above is stated for weak rationalization.

Example 2.8. Let $X = \{a, b, c\}$ and suppose $a \succ^D b$, $b \succ^D c$, and $c \succ^D a$. This strict cycle violates GARP and cannot be generated by a transitive preference relation.

The separation between the abstract choice language and the budget-set language is deliberate. In the next chapter, prices and wealth impose additional structure, and the Afriat inequalities provide a finite certificate that can be checked mechanically.

2.6 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 2.1 [Core]. Incomplete preferences and multiple utilities.

Let $X = \{a, b, c\}$ and let R contain all reflexive pairs and the comparison aRb , but no other comparison between distinct alternatives.

- (a) Show that R is a preorder but is incomplete.
- (b) Prove that no real-valued utility function can represent R in the sense xRy if and only if $u(x) \geq u(y)$.
- (c) Construct two utility functions u_1, u_2 such that xRy if and only if $u_k(x) \geq u_k(y)$ for both $k = 1, 2$.

Problem 2.2 [Core]. Choice cycles on finite menus.

Let $X = \{a, b, c\}$ and suppose a single-valued choice function satisfies $c(\{a, b\}) = a$, $c(\{b, c\}) = b$, and $c(\{a, c\}) = c$. Say that a preference relation R rationalizes these data when the observed alternative is the unique R -greatest element of each observed menu.

- (a) Write the direct revealed-preference comparisons generated by these choices.
- (b) Show that no transitive preference relation can rationalize the data.
- (c) State the smallest change to one observation that restores acyclicity, and exhibit a complete and transitive rationalization of the revised data.

Problem 2.3 [Proof]. Maximization and set-valued WARP.

Let R be complete and transitive on a finite set X . For each nonempty menu $A \subseteq X$, let $c(A)$ be the set of R -greatest elements of A .

- (a) Prove that $c(A)$ is nonempty.
- (b) Suppose $x \in c(A)$, $y \in c(B)$, and $x, y \in A \cap B$. Prove that xIy , where I is the symmetric part of R .
- (c) Explain why this conclusion is the appropriate set-valued version of WARP.

Chapter 3

Consumer Theory: Utility Maximization

This chapter derives demand from preference maximization and develops the duality between utility maximization and expenditure minimization. Throughout, $X = \mathbb{R}_+^L$, $L \geq 1$ is the fixed number of commodities, $P = \mathbb{R}_{++}^L$, $W = \mathbb{R}_{++}$.

3.1 Demand Correspondences

Definition 3.1. A Walrasian demand correspondence is a map

$$x : P \times W \rightrightarrows X$$

with nonempty values such that every $q \in x(p, w)$ satisfies $p \cdot q \leq w$. If demand is generated by a choice correspondence c , then $x = c \circ B$, where

$$B(p, w) = \{q \in X : p \cdot q \leq w\}.$$

Example 3.2. With two perfect-substitute goods and utility $u(q) = q_1 + q_2$,

$$x(p, w) = \begin{cases} \{(w/p_1, 0)\}, & p_1 < p_2, \\ \{(0, w/p_2)\}, & p_2 < p_1, \\ \{q \in \mathbb{R}_+^2 : p_1 q_1 + p_2 q_2 = w\}, & p_1 = p_2. \end{cases}$$

The equal-price value is the entire budget line, not a pair of independent intervals.

For later use, distinguish the following properties.

- A correspondence has a closed graph if $(p^n, w^n) \rightarrow (p, w)$, $q^n \in x(p^n, w^n)$, and $q^n \rightarrow q$ imply $q \in x(p, w)$. Closed graph and upper hemicontinuity are not equivalent without additional compactness or local-boundedness conditions.
- It is compact-valued if $x(p, w)$ is compact for every (p, w) . In Euclidean space, compactness is equivalent to being closed and bounded.
- It is convex valued if $\alpha q + (1 - \alpha)r \in x(p, w)$ whenever $q, r \in x(p, w)$ and $\alpha \in [0, 1]$.

3.2 Utility Representation

Definition 3.3. A preference relation $\succeq$ on $X \subseteq \mathbb{R}^L$ is continuous if its upper and lower contour sets are closed: for every $y \in X$,

$$\{x \in X : x \succeq y\} \quad \text{and} \quad \{x \in X : y \succeq x\}$$

are closed in X . Equivalently, if $x^n \rightarrow x$, then $x^n \succeq y$ for every n implies $x \succeq y$, and $y \succeq x^n$ for every n implies $y \succeq x$.

Lemma 3.4. *Every total preorder on a countable set has a real-valued utility representation.*

Proof. Let I be the symmetric part of the preorder. The quotient X/I is countable, and the relation induced on it is a linear order by Appendix A. Every countable linear order admits an order embedding into $\mathbb{Q}$: enumerate its elements and insert the image of each new element at a rational number between the images of its already embedded lower and upper neighbors. Composing this embedding with the quotient map gives $u : X \rightarrow \mathbb{Q} \subseteq \mathbb{R}$ such that $x \succeq y \iff u(x) \geq u(y)$. $\square$

Theorem 3.5. (*Continuous utility representation*) *If $\succeq$ is a continuous total preorder on $X = \mathbb{R}_+^L$, then there exists a continuous function $u : X \rightarrow \mathbb{R}$ such that, for every $x, y \in X$,*

$$x \succeq y \iff u(x) \geq u(y).$$

Proof. This is Debreu's continuous-utility theorem specialized to Euclidean commodity space. The substantive topological input is that $\mathbb{R}_+^L$ is second countable and connected. Continuity of the preorder and the gap lemma provide a countable order-dense family of indifference classes.

Mapping those classes into a countable dense subset of $\mathbb{R}$ and extending across the gaps yields an order-preserving continuous map. The result cannot be obtained by taking arbitrary limits of a utility defined only on $\mathbb{Q}^L$: such limits need not exist or be independent of the approximating sequence. $\square$

The theorem guarantees existence, not uniqueness. If u represents $\succeq$ and f is strictly increasing on the range of u , then $f \circ u$ represents the same preference relation.

3.3 The Utility Maximization Problem

We assume that u is continuous and $X = \mathbb{R}_+^L$.

The utility maximization problem (UMP) is:

$$\begin{aligned} & \max_{x \geq 0} u(x) \\ & \text{subject to } p \cdot x \leq w. \end{aligned}$$

Equivalently, define the budget set as $x \in B(p, w) = \{x \in \mathbb{R}_+^L : p \cdot x \leq w\}$.

Proposition 3.6. *If $p \gg 0$ and u is continuous, then the UMP has a solution.*

Proof. Because $B(p, w)$ contains the zero bundle, it is nonempty. For each $x \in B(p, w)$ and each commodity ℓ , strict positivity of prices gives $0 \leq x_\ell \leq w/p_\ell$; hence the budget set is bounded.

The map $x \mapsto p \cdot x$ is continuous, so $B(p, w) = \mathbb{R}_+^L \cap \{x : p \cdot x \leq w\}$ is closed. The Heine--Borel theorem therefore implies that $B(p, w)$ is compact.

The Weierstrass maximum theorem therefore gives a solution to the UMP. $\square$

We now prove further properties of the demand correspondence.

Proposition 3.7. *The Marshallian demand correspondence $x(p, w) \equiv \arg \max_{y \in B(p, w)} u(y)$ is non-empty, compact-valued and upper hemicontinuous. Moreover, the indirect utility function $v(p, w) = \max_{y \in B(p, w)} u(y)$ is continuous.*

Proof. The result follows from Berge's maximum theorem once we verify that the budget correspondence $B(\cdot)$ is continuous. We already know that u is continuous and that $B(p, w)$ is compact, so $B(\cdot)$ is compact-valued.

Let $(p^n, w^n) \rightarrow (p, w)$. If $x^n \in B(p^n, w^n)$ and $x^n \rightarrow x$, then $p^n \cdot x^n \leq w^n$; taking limits gives $p \cdot x \leq w$, so the graph is closed.

For the reverse inclusion, let $x \in B(p, w)$. If $p \cdot x < w$, then $x \in B(p^n, w^n)$ for all sufficiently large n . If $p \cdot x = w$, set

$$x^n = \frac{x}{\max\{1, (p^n \cdot x)/w^n\}}.$$

Then $x^n \in B(p^n, w^n)$ and $x^n \rightarrow x$. Hence the budget correspondence is continuous on $\mathbb{R}_{++}^L \times \mathbb{R}_{++}$, and Berge's theorem gives the stated properties. □

We need some definitions.

Definition 3.8. Local nonsatiation. A preference relation $\succeq$ is locally nonsatiated if, for every $x \in X$ and every $\varepsilon > 0$, there is $y \in X$ with $\|y - x\| < \varepsilon$ and $y \succ x$.

Proposition 3.9. *If a locally nonsatiated utility function generates the Marshallian demand correspondence, then every demanded bundle exhausts the budget: $p \cdot x = w$ for every $x \in x(p, w)$.*

Proof. If $p \cdot x < w$, local nonsatiation supplies a strictly preferred bundle arbitrarily close to x . For a sufficiently small neighborhood the bundle remains affordable, contradicting utility maximization. Therefore the budget constraint binds.

Proposition 3.10. *The same demand correspondence is homogeneous of degree zero. If preferences are convex, demand is convex-valued; if preferences are strictly convex, demand is single-valued.*

3.4 KKT Conditions for the UMP

Assume that u is continuously differentiable and that a constraint qualification holds at the candidate optimum. The KKT conditions are then necessary. If u is concave, they are also sufficient for a global optimum; differentiability alone does not provide sufficiency.

The KKT conditions are as follows.

$$\mathcal{L} = u(x) + \lambda(w - p \cdot x)$$

$$\partial_{x_\ell} u(x^*) \leq \lambda^* p_\ell \text{ and } x_\ell^* [\partial_{x_\ell} u(x^*) - \lambda^* p_\ell] = 0 \text{ for all } \ell \in \{1, \dots, L\}.$$

$$w - p \cdot x^* \geq 0 \text{ and } \lambda^* (w - p \cdot x^*) = 0.$$

This is a system with $L + 1$ unknowns.

Local nonsatiation implies that the consumer is on the budget hyperplane (the frontier). This means that

$$w - p \cdot x^* = 0.$$

For an interior solution with a positive multiplier and nonzero marginal utilities, the KKT conditions reduce to

$$\frac{\partial_{x_\ell} u(x^*)}{\partial_{x_k} u(x^*)} = \frac{p_\ell}{p_k} \quad \forall \ell, k \in \{1, \dots, L\}.$$

Note that for $L = 2$, $-\frac{p_1}{p_2}$ is the slope of the budget line and $-\frac{\partial_{x_1} u(x^*)}{\partial_{x_2} u(x^*)}$ is the slope of the indifference curve. Their equality is the familiar tangency condition expressed in terms of the marginal rate of substitution (MRS).

3.5 Properties of the Indirect Utility Function

The indirect utility function records the maximum utility attainable from a budget:

$$v(p, w) = \max\{u(x) : x \in X, p \cdot x \leq w\}.$$

Thus $v : P \times W \rightarrow \mathbb{R}$ ranks price--wealth situations through the utility they permit.

Alternatively it is the utility evaluated in the Marshallian demand $v(p, w) = u(x(p, w))$.

Proposition 3.11. *Suppose that $u(\cdot)$ is a continuous utility function representing a locally nonsatiated preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}_+^L$. The indirect utility function $v(p, w)$ is:*

1. *Homogeneity of degree zero.*
2. *$v(p, w)$ is strictly increasing in w and nonincreasing in p_ℓ .*
3. *$v(p, w)$ is quasiconvex in (p, w) . The set $\{(p, w) | v(p, w) \leq \bar{v}\}$ is convex.*
4. *$v(p, w)$ is continuous in p and w .*

Proof. Existence, homogeneity, monotonicity, and continuity follow from the compact budget correspondence and Berge's maximum theorem.

For quasiconvexity, fix $\bar{v}$ and let $L_{\bar{v}} = \{(p, w) : v(p, w) \leq \bar{v}\}$. If $u(x) > \bar{v}$, then every $(p, w) \in L_{\bar{v}}$ must satisfy $p \cdot x > w$, because otherwise x would be affordable and the value would exceed $\bar{v}$. Conversely, this strict inequality for every x with $u(x) > \bar{v}$ rules out an affordable bundle with utility above $\bar{v}$. Hence

$$L_{\bar{v}} = \bigcap_{u(x) > \bar{v}} \{(p, w) : p \cdot x > w\}.$$

Each set in the intersection is a linear half-space in (p, w) , so the lower contour set is convex and v is quasiconvex.

3.6 Roy's Identity

Recall the definition of indirect utility $v(p, w) = u(x(p, w))$.

Assume the maximizer is unique and the value function is differentiable. The envelope theorem applied to

$$v(p, w) = \max_{x \geq 0, p \cdot x \leq w} u(x)$$

gives

$$\partial_w v(p, w) = \lambda(p, w), \quad \nabla_p v(p, w) = -\lambda(p, w)x(p, w).$$

If demand is unique, the envelope derivatives exist, and $\partial_w v(p, w) = \lambda(p, w) > 0$, Roy's identity is

$$x(p, w) = -\frac{\nabla_p v(p, w)}{\partial_w v(p, w)}.$$

Example 3.12. Quasilinear demand and Roy's identity. Let good L be the numeraire, set $p_L = 1$, and let utility be $u(q, q_L) = g(q) + q_L$ for $q \in \mathbb{R}_+^{L-1}$. Feasibility of the nonnegative numeraire requires $p_{-L} \cdot q \leq w$. The indirect utility is

$$v(p, w) = w + \max_{q \geq 0, p_{-L} \cdot q \leq w} \{g(q) - p_{-L} \cdot q\}.$$

When the unconstrained maximizer of $g(q) - p_{-L} \cdot q$ is affordable and leaves a nonnegative numeraire, the wealth constraint inside the displayed maximum is slack and non-numeraire demands have no income effects. When the maximizer is unique and the envelope theorem applies, Roy's identity gives $x_k(p, w) = -\partial_{p_k} v(p, w)$ for $k < L$, and the numeraire demand is the residual $x_L = w - p_{-L} \cdot x_{-L}(p)$.

3.7 Expenditure Minimization Problem (EMP)

At prices p , the expenditure minimization problem (EMP) asks for the least expenditure required to attain utility $\bar{u}$:

$$e(p, \bar{u}) = \min\{p \cdot x : x \in X, u(x) \geq \bar{u}\}.$$

Proposition 3.13. *Suppose u is a continuous utility function representing a locally nonsatiated preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}_+^L$ and the price vector is $p \gg 0$. Then:*

(i) *If x^* is optimal in the UMP when $w > 0$ then x^* is optimal in the EMP when the required level of utility is $u(x^*)$. Moreover, the minimized expenditure level $p \cdot x^* = w$ is exactly the wealth w .*

(ii) *If x^* is optimal in the EMP when the required utility level is $\bar{u} > u(0)$ then x^* is optimal in the UMP when wealth is $p \cdot x^*$. Moreover, the maximized utility level $u(x^*) = \bar{u}$.*

The proof uses local nonsatiation to show that both constraints bind and then applies the definitions of the two optimization problems.

Expenditure function. The function $e(p, \bar{u})$ is the value of the EMP. Whenever a solution exists, $e(p, \bar{u}) = p \cdot x^*$ for every expenditure-minimizing bundle x^* .

Proposition 3.14. *Suppose that u is a continuous utility function representing a locally non-satiated preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}_+^L$, with $p \in \mathbb{R}_{++}^L$ and the target utility in the attainable range. The expenditure $e(p, \bar{u})$ is*

(i) *Homogeneous of degree one in p .*

(ii) *Nondecreasing in $\bar{u}$ and nondecreasing in p_ℓ for all ℓ . Strict increase in the utility target requires restricting attention to targets above the baseline utility and conditions ensuring that the minimum reaches the target exactly.*

(iii) *Concave in p*

(iv) *Continuous in $(p, \bar{u})$.*

Proof. (i) Multiplying every price by $\alpha > 0$ does not change the feasible set and multiplies the objective by α ; hence $e(\alpha p, \bar{u}) = \alpha e(p, \bar{u})$. For (ii), increasing the utility target shrinks the feasible set, while increasing a price cannot reduce the expenditure on any fixed nonnegative bundle.

(iii) Let $p^\alpha = \alpha p + (1 - \alpha)p'$ with $\alpha \in [0, 1]$ and choose any $x^\alpha \in h(p^\alpha, \bar{u})$. Feasibility of x^α

at each price vector gives

$$\begin{aligned} e(p^\alpha, \bar{u}) &= \alpha p \cdot x^\alpha + (1 - \alpha)p' \cdot x^\alpha \\ &\geq \alpha e(p, \bar{u}) + (1 - \alpha)e(p', \bar{u}). \end{aligned}$$

Hence the expenditure function is concave in prices.

(iv) Continuity in $(p, \bar{u})$.

Continuity follows from the standard expenditure-function theorem. The unbounded feasible set cannot be inserted directly into Berge's theorem. Instead, around any $(p, \bar{u})$ with strictly positive prices, one first obtains a common expenditure upper bound from one feasible bundle. Strict positivity of nearby prices then confines every optimizer to a common compact box. Berge's theorem applies to this locally compactified problem. Continuity of preferences and local nonsatiation ensure feasibility stability and that an optimum reaches the target utility. $\square$

3.8 Hicksian Compensated Demand

The Hicksian (compensated) demand correspondence is the solution set of the EMP: $h(p, \bar{u}) = \arg \min\{p \cdot x : x \in X, u(x) \geq \bar{u}\}$.

Proposition 3.15. *Suppose $u(\cdot)$ is continuous and represents a locally nonsatiated preference relation $\succeq$, prices satisfy $p \gg 0$, and the attainable target satisfies $\bar{u} > u(0)$. Then $h(p, \bar{u})$ is nonempty and compact-valued and has the following properties:*

- (i) *It is homogeneous of degree zero in p*
- (ii) *It has no excess utility: For $x^* \in h(p, \bar{u})$, $u(x^*) = \bar{u}$*
- (iii) *Convexity and uniqueness: If $\succeq$ is convex, then $h(p, \bar{u})$ is convex; if $\succeq$ is strictly convex, then Hicksian demand is single-valued.*

3.9 KKT Conditions for the EMP

Assume u is continuously differentiable and a constraint qualification holds at the candidate optimum. The KKT conditions below are necessary. If u is concave, they are also sufficient.

The Lagrangian for this problem is:

$$\mathcal{L}_{\text{EMP}} = p \cdot x - \lambda(u(x) - \bar{u}).$$

Stationarity and complementary slackness for the nonnegativity constraints give

$$p \geq \lambda \nabla_x u(x^*)$$

$$x_\ell^* [p_\ell - \lambda \nabla u(x^*)_\ell] = 0,$$

for some $\lambda \geq 0$, together with primal feasibility $x^* \geq 0$ and $u(x^*) \geq \bar{u}$, and complementary slackness $\lambda[u(x^*) - \bar{u}] = 0$.

3.10 Shephard's Lemma

Proposition 3.16. *Every Hicksian bundle is a supergradient of the concave expenditure function in prices. If $e(\cdot, \bar{u})$ is differentiable at p , Hicksian demand is unique there and Shephard's identity holds:*

$$\nabla_p e(p, \bar{u}) = h(p, \bar{u}).$$

Proof. Fix $x^h \in h(p, \bar{u})$. For every other price vector p' , the same bundle remains feasible, so

$$e(p', \bar{u}) \leq p' \cdot x^h = e(p, \bar{u}) + (p' - p) \cdot x^h.$$

Hence x^h is a supergradient of the concave function $e(\cdot, \bar{u})$. At a differentiability point the superdifferential is the singleton $\{\nabla_p e(p, \bar{u})\}$, so every Hicksian bundle equals that gradient and demand is unique. □

3.11 Additional Properties

Assume throughout this subsection that $e(\cdot, \bar{u})$ is twice continuously differentiable, so Hicksian demand is single-valued and differentiable. The Jacobian $D_p h(p, \bar{u})$ has entries $[D_p h(p, \bar{u})]_{k\ell} = \partial_{p_\ell} h_k(p, \bar{u})$.

Proposition 3.17. *The Hicksian price-effect matrix satisfies*

$$D_p h(p, \bar{u}) = D_p^2 e(p, \bar{u}).$$

It is symmetric and negative semidefnite, and $D_p h(p, \bar{u})p = 0$. The first claims follow from

Shephard's identity, twice continuous differentiability, and concavity of the expenditure function. The last follows by differentiating the degree-zero homogeneity of Hicksian demand.

Definition 3.18. Negative semidefiniteness. A symmetric $L \times L$ matrix A is negative semidefinite if $v^\top A v \leq 0$ for every vector $v \in \mathbb{R}^L$.

Example 3.19. The following matrix is NSD:

$$A = \begin{bmatrix} -2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$v^\top A v = -2v_1^2 \leq 0$$

for any $v \in \mathbb{R}^2$.

3.12 Summary of Relationships between the EMP and UMP

The expenditure function is the value of the EMP: for every $x^h \in h(p, \bar{u})$, $e(p, \bar{u}) = p \cdot x^h$.

Under continuity and local nonsatiation, at positive prices and feasible utility targets, the demand and value functions satisfy the duality identities

$$h(p, v(p, w)) = x(p, w),$$

$$x(p, e(p, \bar{u})) = h(p, \bar{u}),$$

$$v(p, e(p, \bar{u})) = \bar{u},$$

$$e(p, v(p, w)) = w.$$

The first two equalities are equalities of correspondences; when demand is single-valued, the same notation denotes functions.

3.13 Slutsky Equation

Assume in this section that Marshallian and Hicksian demands are single-valued and continuously differentiable and that the expenditure function is differentiable in prices. The Slutsky matrix is $S : P \times W \rightarrow \mathbb{R}^{L \times L}$, $S(p, w)_{\ell, k} = \partial_{p_k} x_\ell(p, w) + \partial_w x_\ell(p, w) x_k(p, w)$, $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$.

Differentiate the duality identity $h(p, \bar{u}) = x(p, e(p, \bar{u}))$ with respect to p :

$$D_p h(p, \bar{u}) = D_p x(p, e(p, \bar{u})) + D_w x(p, e(p, \bar{u})) \nabla_p e(p, \bar{u})^\top$$

By Shephard's lemma and the duality identities,

$$D_p h(p, v(p, w)) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top = S(p, w).$$

3.14 Summary of the Implications of Utility Maximization

Proposition 3.20. *Suppose demand $x(p, w)$ is generated by maximizing a complete, transitive, continuous, and locally nonsatiated preference relation represented by u on each linear budget; that is, $x(p, w) \in \arg \max_{y \in B(p, w)} u(y)$ for every $(p, w) \in P \times W$. Then demand satisfies Walras' law and is homogeneous of degree zero.*

If, in addition, demand is single-valued and continuously differentiable and the expenditure function is twice continuously differentiable, then the Slutsky matrix $S(p, w)$ is symmetric and negative semidefinite, and it is singular in the price direction: $S(p, w)p = 0$ for every $(p, w) \in P \times W$.

This proposition summarizes our previous results.

3.15 Integrability

We now ask for sufficient conditions under which an observed demand function is generated by utility maximization. In the abstract choice setting, WARP and GARP restrict rationalizability. Competitive budget sets are a smaller domain than arbitrary menus, so differentiable demand requires additional restrictions. The relevant restrictions are stated in terms of the Slutsky matrix.

Assume throughout this subsection that the demand function x is continuously differentiable.

Recall that the Slutsky matrix function is $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^\top$, or in the scalar version $s_{ij}(p, w) = \partial_{p_j} x_i(p, w) + \partial_w x_i(p, w) x_j(p, w)$ for all $i, j \in \{1, \dots, L\}$.

Remark 3.21. Integrability roadmap. On an open domain, Walras' law, homogeneity, and a symmetric negative-semidefinite Slutsky matrix are the central differential restrictions. They are necessary for every regular utility-generated demand. A sufficiency theorem additionally specifies rank or nondegeneracy conditions, the topology of the price-wealth domain, and boundary

behavior for the associated differential system. Because alternative integrability theorems impose different global hypotheses, this text does not present the preceding checklist as a stand-alone global theorem; applications must cite and verify a precise version.

The qualifications are essential. Symmetry and negative semidefiniteness alone, stated on an unspecified domain, do not guarantee a global representation: one must control the domain, boundary behavior, and the differential equation's existence and uniqueness. Conversely, every continuously differentiable demand generated by a regular locally nonsatiated utility satisfies Walras' law, homogeneity, symmetry, and negative semidefiniteness of the Slutsky matrix.

Remark. Some textbooks require a boundedness condition on the elasticity of wealth, this is in general not required.

3.16 Examples and Applications of Demand Integrability

There are two commodities $L = 2$ and you estimate a demand function $x_1(p, w) = \frac{\alpha w}{p_1}$, and you know that the demand of good two is given as a residual. Furthermore, we have checked that this demand has a Slutsky matrix that is symmetric, NSD and singular in prices, we normalize $p_2 = 1$. Thus $x_2 = (1 - \alpha)w$.

The $L - 1$ differential equation is:

$$\mu'(p_1, 1) = \frac{\alpha \mu(p_1, 1)}{p_1} = x_1((p_1, 1), \mu((p_1, 1))),$$

for short, we can write:

$$\mu'(p_1) = \frac{\alpha \mu(p_1)}{p_1} = x_1(p_1, \mu(p_1)),$$

We can write more simply:

$$\frac{\mu'}{\mu} = \frac{\alpha}{p_1}.$$

Simple integration provides a family of solutions:

$$\ln \mu = \alpha \ln p_1 + C,$$

and in levels:

$$\mu = K p_1^\alpha,$$

with $K = \exp(C)$ a constant of integration.

Given some initial conditions (p_1^0, w^0) , we must have:

$$w^0 = K (p_1^0)^\alpha, \quad K = \frac{w^0}{(p_1^0)^\alpha},$$

or

$$\mu(p_1^*, p_1^0, w^0) = \frac{w^0}{(p_1^0)^\alpha} (p_1^*)^\alpha.$$

Normalize $p_1^* = 1$ and define the indirect utility by

$$v(p_1, w) = \mu(p^*, p_1, w) = \frac{w}{p_1^\alpha}.$$

To recover the utility u , we need to invert the demand, note that our primitive is indeed invertible.

We have to use the demand function and obtain the inverse demand:

$$x_1(p, w) = \frac{\alpha w}{p_1}; \quad x_2(p, w) = (1 - \alpha)w$$

$$\frac{x_1}{x_2} = \frac{\alpha}{(1 - \alpha)} \frac{1}{p_1}$$

$$p_1 = \frac{\alpha}{1 - \alpha} \frac{x_2}{x_1}, \quad w = \frac{x_2}{1 - \alpha}.$$

Replacing back in the indirect utility we have:

$$u(x_1, x_2) = v \circ (p_1, w)(x_1, x_2) = v\left(\frac{\alpha}{1 - \alpha} \frac{x_2}{x_1}, \frac{x_2}{1 - \alpha}\right).$$

$$u(x_1, x_2) = \frac{\frac{x_2}{1 - \alpha}}{\left(\frac{\alpha}{1 - \alpha} \frac{x_2}{x_1}\right)^\alpha} = \left(\frac{x_2}{1 - \alpha}\right)^{1 - \alpha} \left(\frac{x_1}{\alpha}\right)^\alpha$$

$$u(x_1, x_2) = \hat{c} x_1^\alpha x_2^{1 - \alpha}, \quad \hat{c} = \frac{1}{\alpha^\alpha (1 - \alpha)^{1 - \alpha}}.$$

This is a Cobb-Douglas utility function.

3.17 Generalized Axiom of Revealed Preference (GARP)

We now move from unrestricted demand functions to finite observations. Suppose that the dataset is $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$, $T \geq 2$. Walras' law is maintained throughout this section.

Definition 3.22. The observed bundle x^t is directly revealed preferred to x , written $x^t R^D x$, if and only if $p^t \cdot x^t \geq p^t \cdot x$.

The observed bundle x^t is indirectly revealed preferred to x , written $x^t R x$, if and only if there exist $m \geq 1$ and bundles $z^0, \dots, z^m$ such that $z^0 = x^t$, $z^m = x$, each z^{k-1} is an observed bundle, and $z^{k-1} R^D z^k$ for every $k = 1, \dots, m$. Thus R is the transitive closure of R^D on the observed data.

Utility maximization can be set-valued, so repeated observations of the same budget may contain different maximizing bundles. The relevant empirical notion is therefore weak rationalization.

Definition 3.23. We say that a utility function u (weakly) rationalizes a dataset $\mathcal{O}$ if, for every $t \in \{1, \dots, T\}$ and every $x \in X$, $p^t \cdot x \leq p^t \cdot x^t$ implies $u(x^t) \geq u(x)$.

We now introduce a property that serves as a test of rationalizability. We can define the strict direct preference relation as $x^t P^D x^s$ if and only if $p^t \cdot x^t > p^t \cdot x^s$.

Definition 3.24. A finite demand dataset $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ satisfies the generalized axiom of revealed preference (GARP) if

$$x^t R x^s \implies \neg(x^s P^D x^t)$$

for all observations s, t , where R is the transitive closure of direct revealed preference and

$$x^s P^D x^t \iff p^s \cdot x^s > p^s \cdot x^t.$$

Equivalently, a chain of weak revealed-preference comparisons may not be closed by a strict direct comparison in the opposite direction.

Proposition 3.25. (*Afriat's theorem.*) Let $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ be a finite dataset with $p^t \gg 0$ and $x^t \in \mathbb{R}_+^L$ for every observation t . The following are equivalent:

1. There exists a locally nonsatiated utility function u that rationalizes the data in the sense that for all t , $u(x^t) \geq u(x)$ for all x such that $p^t \cdot x^t \geq p^t \cdot x$.
2. The dataset $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ satisfies GARP.
3. There exist numbers $u^t \in \mathbb{R}$ and $\lambda^t > 0$ satisfying the Afriat inequalities

$$u^t \leq u^s + \lambda^s p^s \cdot (x^t - x^s) \quad \forall s, t \in \{1, \dots, T\}$$

4. There exists a continuous, concave, and strictly increasing utility function u that rationalizes the data in the sense that for all t , $u(x^t) \geq u(x)$ for all x such that $p^t \cdot x^t \geq p^t \cdot x$.

Proof. (1) $\implies$ (2), (2) $\implies$ (3), (3) $\implies$ (4), (4) $\implies$ (1).

(1) implies (2),

Assume that $\mathcal{O}$ is rationalized by a locally nonsatiated utility function and suppose, toward a contradiction, that GARP fails. Then there are observation indices $t_0, \dots, t_m$ such that $x^{t_{k-1}} R^D x^{t_k}$ for $k = 1, \dots, m$ and $x^{t_m} P^D x^{t_0}$. Rationalization gives $u(x^{t_0}) \geq \dots \geq u(x^{t_m})$. The strict comparison implies $p^{t_m} \cdot x^{t_0} < p^{t_m} \cdot x^{t_m}$. By local nonsatiation, there is a bundle y with $\|y - x^{t_0}\| < \varepsilon$ and $u(y) > u(x^{t_0})$; choosing $\varepsilon > 0$ sufficiently small preserves strict affordability at prices p^{t_m} . Thus y is affordable at observation t_m and satisfies $u(y) > u(x^{t_m})$, contradicting rationalization.

(2) implies (3)

The implication from GARP to the Afriat inequalities is the finite Afriat feasibility lemma. It is a theorem of alternatives for the displayed linear system, not an immediate consequence of naming Farkas' lemma. We cite the lemma here: for a finite, expenditure-exhausting dataset with strictly positive prices, GARP holds if and only if there exist $(u^1, \dots, u^T)$ and positive $(\lambda^1, \dots, \lambda^T)$ satisfying every Afriat inequality. The converse can be seen directly by summing the inequalities around a revealed-preference cycle; the forward direction uses the full finite feasibility lemma.

(3) implies (4)

Let $u(x) = \min_{t \in \{1, \dots, T\}} \{u^t + \lambda^t p^t \cdot (x - x^t)\}$. As the pointwise minimum of finitely many affine functions, u is continuous and concave. It is strictly increasing in the product order because $\lambda^t > 0$ and $p^t \gg 0$ for every observation t .

To verify rationalization, first observe that $u(x^t) = u^t$ for every observation t . Indeed, the Afriat inequality gives

$$u^t \leq u^m + \lambda^m p^m \cdot (x^t - x^m)$$

for every m , while the term with $m = t$ equals u^t . Therefore the minimum of these affine terms is exactly u^t .

It remains to show that $u(x) \leq u(x^s)$, if $p^s \cdot x^s \geq p^s \cdot x$

If $p^s \cdot x^s \geq p^s \cdot x$, or $p^s \cdot (x^s - x) \geq 0$, then $u(x) = \min_t \{u^t + \lambda^t p^t \cdot (x - x^t)\} \leq u^s + \lambda^s p^s \cdot (x - x^s) \leq u^s = u(x^s)$.

Thus $u(x^s) \geq u(x)$ when $p^s \cdot x^s \geq p^s \cdot x$.

(4) implies (1).

If u is strictly increasing, then it is locally nonsatiated.

□

3.18 Revealed Preference and the Strong Axiom

The preceding section used GARP to characterize weak rationalization, allowing ties in observed choice. This section records the corresponding strict result: with single-valued choice and strict concavity, SARP replaces GARP and the Afriat inequalities become strict. The two results are parallel but answer different rationalizability questions.

Definition 3.26. (Strong axiom of revealed preference, SARP.) The dataset satisfies SARP if the transitive closure R has no nontrivial cycle: whenever $x^t R x^s$ and $x^t \neq x^s$, it is not the case that $x^s R x^t$.

SARP is the natural condition when observed choice is single-valued and ties are ruled out. The weak Afriat theorem above is the maintained finite-data result in this book; strict rationalization results require additional assumptions, including care with duplicate observed bundles.

3.19 Forecasting with Varian's Support Set

Definition 3.27. Given a finite price-choice dataset $\mathcal{O}$ and a new price-wealth pair (p^{K+1}, w^{K+1}) , define the augmented dataset associated with a candidate bundle by $\mathcal{O}(x) = \mathcal{O} \cup \{(p^{K+1}, x)\}$. Varian's support set is

$$S(p^{K+1}, w^{K+1}) = \left\{ x \in X : \begin{array}{l} \mathcal{O}(x) \text{ satisfies GARP,} \\ p^{K+1} \cdot x = w^{K+1} \end{array} \right\}.$$

The definition already gives the exact criterion: a candidate bundle belongs to the support set if it exhausts the new budget and the augmented dataset satisfies GARP. The transitive closure must be checked, not only one-step comparisons. Whenever an old bundle x^t is indirectly revealed preferred to the candidate, GARP requires the inequality

$$p^{K+1} \cdot x \leq p^{K+1} \cdot x^t.$$

3.20 Upper Bounds for Welfare Analysis

Definition 3.28. A money metric utility function is defined as $m(p, x) = e(p, u(x))$ where e is the expenditure function associated with u , $e(p, \bar{u}) = \min_{z \in X} \{p \cdot z : u(z) \geq \bar{u}\}$.

For any given p , $m(p, x)$ is a utility function that represents the same preferences as u but is expressed in money units.

Given a finite dataset $\mathcal{O}$, Varian proposes the following upper bound for money-metric utility:

$$m^+(p, x) = \min\{p \cdot x^t : x^t R x, t = 1, \dots, T\}.$$

3.21 Experiments about Testing Rationality

(This material is presented in Quah¹)

- Data from the portfolio choice experiment in Choi, Fisman, Gale, and Kariv (AER, 2007).
- 93 undergraduate subjects participated in the experiment at UC Berkeley, each completing 50 decision problems under risk.
- There were two states of the world, each occurring with a known probability, and two Arrow-Debreu securities, one for each state.
- In each decision problem, every subject was given a budget; income was normalized to one and state prices were chosen at random. 47 subjects were subjected to the symmetric treatment, where $\pi_1 = \pi_2 = \frac{1}{2}$. The rest had the asymmetric treatment where $\pi_1 = \frac{1}{3}$, $\pi_2 = \frac{2}{3}$ (or vice versa).

¹<http://www.johnquah.com/uploads/7/5/6/7/75676677/survey-qutweb.pdf>

3.22 Results

- In the symmetric treatment, 12/47 pass GARP.
- In the asymmetric treatment, 4/46 pass GARP.
- Roughly 17% pass GARP.

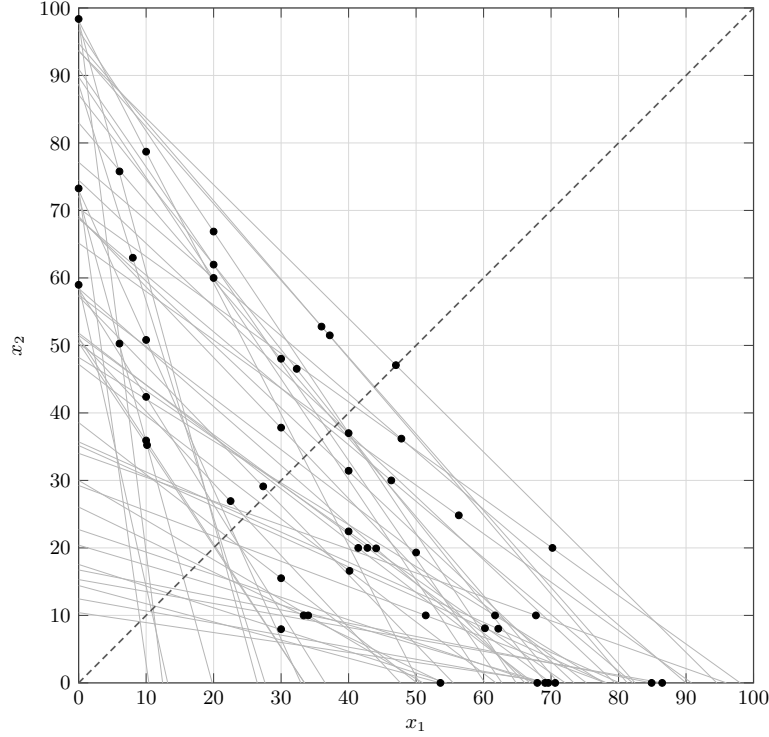


Figure 3.1: Revealed-preference choices of Subject 216 in the symmetric treatment. Gray lines are observed budgets, black dots are chosen bundles, and the dashed line is the 45-degree line. Coordinates recovered from Quah's survey slides; underlying experiment: [Quah, 2015, Choi et al., 2007].

3.23 Afriat's Cost-Efficiency Index

When a dataset $\mathcal{O}$ fails GARP, the critical cost-efficiency index measures the size of the revealed-preference violation. For an efficiency level $\alpha \in [0, 1]$, define the efficiency-adjusted direct relations by

$$x^t R_\alpha^D x^s \iff \alpha p^t \cdot x^t \geq p^t \cdot x^s, \quad x^t P_\alpha^D x^s \iff \alpha p^t \cdot x^t > p^t \cdot x^s.$$

Let R_α be the transitive closure of R_α^D . The dataset satisfies α -GARP if, for all observations

$s, t, x^t R_\alpha x^s$ implies $\neg(x^s P_\alpha^D x^t)$.

Definition 3.29. The Critical Cost Efficiency Index (CCEI) of a dataset $\mathcal{O}$ is $\text{CCEI}(\mathcal{O}) = \sup\{\alpha \in [0, 1] : \mathcal{O} \text{ satisfies } \alpha\text{-GARP}\}$. A value of one means that the original data satisfy GARP; a smaller value is the largest common fraction of observed expenditure that can be retained while eliminating every GARP violation.

The CCEI distribution for the experiment is shown below.

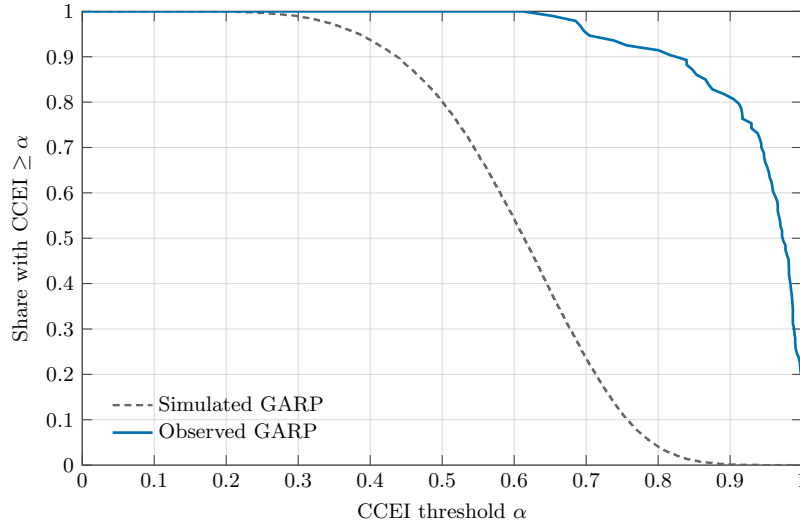


Figure 3.2: Share of observations whose CCEI is at least α : observed GARP compliance and Quah's simulated GARP benchmark. Curves recovered from Quah's survey slides (slide 27) [Quah, 2015].

3.24 Measurement Error

A similar experiment of budget allocation but using 3 goods instead of 2, is in Ahn, D., Choi, S., Gale, D., & Kariv, S. (2014) (Estimating ambiguity aversion in a portfolio choice experiment. Quantitative Economics, 5(2), 195–223). This experiment has $n = 154$ individuals, and $T = 50$ trials.

- The GARP test is successful 12.99% of cases (individuals).
- Aguiar and Kashaev (2020, Restud) extend GARP to allow for measurement error.
- Measurement error can be either present in consumption $x_t = x_t^* + w_t^c$ or in prices $p_t = p_t^* + w_t^p$ for x_t^*, p_t^* being the correctly measured prices and consumption.

- GARP should hold for the (ideal), well-measured dataset $(p_t^*, x_t^*)_{t=1}^T$ but not necessarily for the mismeasured dataset $(p_t, x_t)_{t=1}^T$.
- Without any knowledge about measurement error then rationality becomes untestable.
- We assume that measurement error is random across individuals and that the expectation is equal to zero: $\mathbb{E}[\mathbf{w}_t^c] = 0$, $\mathbb{E}[\mathbf{w}_t^p] = 0$.
- These assumptions are backed up by validation experimental data that shows that consumers may incur in trembling hand errors, or perception errors that are not systematic (see our paper).
- The key is that we do not know the distribution of measurement error, we only know it's average behavior.
- Aguiar and Kashaev (2020) provide a methodology to test GARP in the ideal (correctly measured data) with knowledge about average behavior of measurement error.
- They conclude that the null hypothesis that every individual in this experiment is consistent with GARP cannot be rejected at the 95 confidence level. They also find that misperception of prices seems to be the culprit of spurious violations of rationality.
- Does this confirm rationality? NO!
- A theory cannot be confirmed by this exercise alone, but current evidence is insufficient to reject rationality (at least in the context of this simple experiment).

3.25 Welfare Evaluation

We will compare the wealth needed to achieve a certain level of utility at a given price-wealth pair.

$$u^1 = v(p^1, w)$$

$$u^0 = v(p^0, w).$$

We are interested in comparing the expenditure needed to achieve u^1 or u^0 .

For a fixed reference price and a given utility level u , the needed expenditure to reach it is given by

$$e(p^{ref}, u).$$

Thus the comparison in terms of expenditure is given by

$$e(p^{ref}, u^1) - e(p^{ref}, u^0)$$

$$e(p^{ref}, v(p^1, w)) - e(p^{ref}, v(p^0, w)).$$

This expression will be positive when it takes more wealth to achieve u^1 at prices p^{ref} than to achieve u^0 . This difference is measured in dollar terms. That is why $e(p^{ref}, v(p, w))$ is often called money metric indirect utility function.

We can construct many money metric indirect utility functions with different reference price.

When $p^{ref} = p^0$ is the original price situation, the expenditure change equals the wealth change that makes the consumer indifferent between the new price with the old wealth and the old price with the new wealth. This asks what wealth change is equivalent to the price change.

Equivalent variation is

$$EV(p^0, p^1, w) = e(p^0, v(p^1, w)) - e(p^0, v(p^0, w)) = e(p^0, v(p^1, w)) - w,$$

since $e(p^0, v(p^0, w)) = w$.

With this sign convention, equivalent variation is the money change at the reference price that is equivalent to the price change: it is willingness to pay for an improvement, and its negative is willingness to pay to avoid a worsening.

The other case considered is the one where the new price is taken as the reference price, when $p^{ref} = p^1$, the expenditure change equals the wealth change that makes the consumer indifferent between the original situation (p^0, w) and the new situation $(p^1, w + \Delta w)$.

Compensating variation is

$$CV(p^0, p^1, w) = e(p^1, v(p^1, w)) - e(p^1, v(p^0, w)) = w - e(p^1, v(p^0, w)).$$

The sign convention for compensating variation is analogous: a negative value for a worsening is the compensation required to restore the initial utility.

3.26 Homothetic and Quasilinear Utility Functions

Definition 3.30. Homothetic preferences. A preference relation is homothetic if $x \succeq y$ implies $\alpha x \succeq \alpha y$ for every $\alpha > 0$ whenever the scaled bundles are feasible. Under the usual continuity and monotonicity assumptions, it admits a positively homogeneous utility representation; an arbitrary representation need not itself be homogeneous.

The Cobb–Douglas representation $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ is a standard example.

Definition 3.31. Quasilinear preferences. A numeraire good L is quasilinear when preferences admit a representation

$$u(x) = g(x_1, \dots, x_{L-1}) + x_L.$$

Equivalently, translating both bundles by the same amount of the numeraire does not change their ranking whenever both translated bundles remain feasible.

3.27 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 3.1 [Core]. Cobb–Douglas duality.

For $u(x) = x_1^\alpha x_2^{1-\alpha}$ with $0 < \alpha < 1$, $p \gg 0$, $w > 0$, and $\bar{u} > 0$, derive the Marshallian demand $x(p, w)$, indirect utility $v(p, w)$, Hicksian demand $h(p, \bar{u})$, and expenditure function $e(p, \bar{u})$.

- (a) Verify Walras' law and the identity $e(p, v(p, w)) = w$.
- (b) Verify $h(p, v(p, w)) = x(p, w)$.
- (c) Verify Roy's identity and Shephard's lemma.

Problem 3.2 [Core]. Recovering a quasilinear utility.

Normalize $p_2 = 1$ and suppose an interior demand system satisfies $x_1(p_1, w) = \beta - \alpha p_1$, where $\alpha, \beta > 0$, and Walras' law.

- (a) Derive $x_2(p_1, w)$ and state the domain on which both demands are nonnegative.
- (b) Compute the normalized scalar Slutsky term for the non-numeraire demand.
- (c) Recover a quasilinear utility representation $u(x_1, x_2) = g(x_1) + x_2$ that generates the demand on the interior domain. Distinguish general ordinal uniqueness up to an increasing transformation from the narrower normalization freedom after the coefficient on x_2 is fixed at one.

Problem 3.3 [Core]. Afriat inequalities by hand.

Consider two observations: $p^1 = (1, 1)$, $x^1 = (2, 1)$ and $p^2 = (2, 1)$, $x^2 = (1, 1)$.

- (a) Compute observed expenditures and all direct revealed-preference comparisons.
- (b) Check GARP.
- (c) Verify that $u^1 = 1$, $u^2 = 0$, and $\lambda^1 = \lambda^2 = 1$ satisfy the Afriat inequalities.
- (d) Write the corresponding piecewise-linear Afriat utility and verify that it assigns the required utility levels at the observations.

Problem 3.4 [Proof]. The Slutsky matrix for Cobb-Douglas demand.

Using the Marshallian demand from Problem 3.1, compute $S(p, w) = D_p x + D_w x x^\top$.

- (a) Show that S is symmetric.
- (b) Show directly that $z^\top S z \leq 0$ for every $z \in \mathbb{R}^2$.
- (c) Find the null space of S and interpret it using homogeneity.

Problem 3.5 [Applied]. Demand forecasting with Varian's support set.

A finite dataset $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^T$ satisfies GARP. Fix a counterfactual budget (p^0, w^0) with $p^0 \gg 0$, and define $S(p^0, w^0) = \{x^0 : p^0 \cdot x^0 = w^0, \mathcal{O} \cup \{(p^0, x^0)\} \text{ satisfies GARP}\}$.

- (a) State a revealed-preference condition under which the data imply $x^t R x^0$.
- (b) Derive the associated linear restriction on $p^0 \cdot x^0$.
- (c) Explain why the restrictions in parts (a)–(b) are necessary but must be combined with every revealed-preference path into and out of the candidate observation to characterize $S(p^0, w^0)$. Explain why this set gives bounds rather than point identification.

Part II

Choice Beyond the Benchmark Model

Chapter 4

Behavioral Economics: Sparse-Max and Rational Inattention

The benchmark consumer maximizes utility using the true budget and prices. A sparse-max consumer keeps the true budget constraint but forms a decision using a perceived price vector. This chapter states the behavioral rule explicitly and records which benchmark conclusions no longer follow automatically.

4.1 The Sparse-Max Decision Rule

Let $X = \mathbb{R}_+^L$, true prices $p \gg 0$, perceived prices $\hat{p} \gg 0$, and wealth $w > 0$. Assume u is differentiable and that the tangency system below has a solution. A sparse-max choice is obtained by first selecting a bundle that satisfies the perceived-price first-order condition

$$\nabla u(x) = \lambda \hat{p}, \quad \lambda > 0,$$

and then choosing λ so that the selected bundle satisfies the actual budget equation $p \cdot x = w$. At a boundary solution, replace the equality by the corresponding KKT inequalities. This two-stage rule is written $x^s(p, \hat{p}, w)$. It is not the ordinary utility-maximization problem with $\hat{p}$ in the budget constraint.

When the rational demand $x^r(\hat{p}, \tilde{w})$ is single-valued and locally invertible in wealth, the

same sparse-max bundle can be written as $x^s(p, \hat{p}, w) = x^r(\hat{p}, \tilde{w})$ for the unique $\tilde{w}$ satisfying $p \cdot x^r(\hat{p}, \tilde{w}) = w$. This is a change of representation, not an assertion that the consumer faces $\hat{p}$ in the market.

4.2 Cobb--Douglas Sparse-Max Demand

Example 4.1. Let $u(x) = \sum_{\ell=1}^L \alpha_{\ell} \log x_{\ell}$ with $\alpha_{\ell} > 0$ and $\sum_{\ell} \alpha_{\ell} = 1$. The perceived-price first-order conditions give

$$x_{\ell} = \frac{\alpha_{\ell}}{\lambda \hat{p}_{\ell}}.$$

Imposing the actual budget yields $\lambda = \frac{1}{w} \sum_{k=1}^L \alpha_k \frac{p_k}{\hat{p}_k}$ and therefore

Example 4.2.

$$x_{\ell}^s(p, \hat{p}, w) = \frac{\alpha_{\ell} w / \hat{p}_{\ell}}{\sum_{k=1}^L \alpha_k p_k / \hat{p}_k}.$$

Indeed, $\sum_{\ell} p_{\ell} x_{\ell}^s = w$. If $\hat{p} = p$, this reduces to the ordinary Cobb--Douglas demand $x_{\ell} = \alpha_{\ell} w / p_{\ell}$.

4.3 Perceived Prices and Limited Attention

One parsimonious attention rule combines the true price with a default price $p^d \gg 0$:

$$\hat{p}_{\ell}(p) = m_{\ell} p_{\ell} + (1 - m_{\ell}) p_{\ell}^d, \quad m_{\ell} \in [0, 1].$$

The vector $m = (m_1, \dots, m_L)$ measures attention to price components. Full attention is $m_{\ell} = 1$ for every ℓ ; setting $m_{\ell} = 0$ makes that component equal to its default. With $M = \text{diag}(m_1, \dots, m_L)$, $\hat{p}(p) = Mp + (I - M)p^d$.

Proposition 4.3. *Comparative statics under perceived prices. Suppose the tangency bundle can be written as a differentiable map $r(\hat{p}, \lambda)$ and $x^s(p, w) = r(\hat{p}(p), \lambda(p, w))$, where $\lambda(p, w)$ is determined by $p \cdot r(\hat{p}(p), \lambda) = w$. Then the chain rule gives*

$$D_p x^s = D_{\hat{p}} r M + \partial_{\lambda} r (\nabla_p \lambda)^{\top}, \quad D_w x^s = \partial_{\lambda} r \partial_w \lambda.$$

The second terms are the response of the multiplier needed to restore the actual budget. Consequently, an identity such as $S^s = S^r M$ requires additional compensation and normalization

assumptions; it is not a generic implication of the attention rule.

Because the default price is fixed, scaling (p, w) by a common positive factor generally changes $\hat{p}(p)$ and therefore can violate homogeneity of degree zero. WARP, Slutsky symmetry, and welfare theorems must likewise be checked for the specified behavioral demand rather than imported from the rational benchmark.

4.4 Endogenizing Attention

Let the unknown scalar state be $\theta \sim \mathcal{N}(\theta^d, \sigma_\theta^2)$ and let the signal be $Y = \theta + \varepsilon$ with independent $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. Under quadratic loss $\ell(a, \theta) = \frac{1}{2}(a - \theta)^2$, the optimal action is the posterior mean

$$a^*(y) = \mathbb{E}[\theta \mid Y = y] = my + (1 - m)\theta^d, \quad m = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2} \in [0, 1].$$

The posterior variance is $(1 - m)\sigma_\theta^2$, so expected loss at the optimal action is $\frac{1}{2}(1 - m)\sigma_\theta^2$. If attention itself is costly, represent the cost by an increasing function $C(m)$ and solve

$$\max_{0 \leq m < 1} -\frac{1}{2}(1 - m)\sigma_\theta^2 - C(m).$$

For example, with $C(m) = \kappa m^2/2$, an interior optimum satisfies $m = \sigma_\theta^2/(2\kappa)$ and is truncated to $[0, 1)$ when the unconstrained value lies outside that interval. The comparative statics depend on the cost specification.

A capacity formulation is also transparent. If the information cost is $-\frac{1}{2}\log(1 - m)$ and capacity is bounded by K , then $-\frac{1}{2}\log(1 - m) \leq K$ implies $m \leq 1 - e^{-2K}$. Since the value of information is increasing in m , the capacity-constrained optimum attains this upper bound.

4.5 Discrimination with Rational Inattention

Consider a firm deciding whether to hire an applicant. Productivity is Q and hiring yields payoff Q , while rejecting yields 0. Before observing an applicant-specific signal, the firm observes $G \in \{R, B\}$ and has prior $Q \mid G \sim \mathcal{N}(q_G, \sigma_q^2)$ with $q_R < q_B$.

The firm observes $Y = Q + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is independent. The posterior mean is

$$m_G(y) = \mathbb{E}[Q \mid Y = y, G] = \alpha y + (1 - \alpha)q_G, \quad \alpha = \frac{\sigma_q^2}{\sigma_q^2 + \sigma_\varepsilon^2}.$$

Proposition 4.4. *Optimal hiring rule. Conditional on (y, G) , hire if and only if $m_G(y) > 0$;*

otherwise reject. For $\alpha > 0$ the signal threshold is $y_G = -(1 - \alpha)q_G/\alpha$.

If both group means are negative, the lower-mean group has the higher threshold and is hired less often under a common signal distribution. If both means are positive, the same threshold formula applies, but group-specific rejection rates depend on the signal distribution. If information is chosen, parameterize the decision by precision or by the informativeness index $\alpha \in [0, 1)$ and solve $\max_{0 \leq \alpha < 1} \mathbb{E}[\max\{m_G(Y), 0\}] - C(\alpha)$ with an increasing attention cost C . Greater noise variance means less attention, so a cost increasing in σ_ε^2 would reverse the intended economic tradeoff. Group-specific attention requires the group priors and a fully specified cost function.

4.6 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 4.1 [Core]. Sparse-max quasilinear demand.

Let $u(x_1, x_2) = \sqrt{x_1} + x_2$. The consumer uses perceived prices $\hat{p} \gg 0$ in the first-order condition but must satisfy the actual budget $p \cdot x = w$.

- (a) Derive the interior sparse-max choice.
- (b) State the condition under which the implied x_2 is nonnegative.
- (c) Show that ordinary Marshallian demand is recovered when $\hat{p} = p$.

Problem 4.2 [Core]. CES demand under perceived prices.

Let $u(x) = [\alpha x_1^\rho + (1 - \alpha)x_2^\rho]^{1/\rho}$, where $0 < \alpha < 1$ and $\rho < 1$, $\rho \neq 0$.

- (a) Use perceived prices $\hat{p}$ to derive the chosen ratio x_1/x_2 .
- (b) Scale the bundle so that it exhausts the actual budget $p \cdot x = w$.
- (c) Identify separately the perceived-price composition effect of changing $\hat{p}$ and the feasibility effect of changing p . Explain why the first is not automatically a Hicksian substitution effect.

Problem 4.3 [Applied]. A binary rational-inattention problem.

There are two equally likely states and two actions. The decision maker receives payoff one from the correct action and zero from the incorrect action. A symmetric strategy is correct with probability $q \in [1/2, 1]$ and pays Shannon-information cost $\kappa[\log 2 - H(q)]$, where $H(q) = -q \log q - (1 - q) \log(1 - q)$ and $\kappa > 0$.

- (a) Write expected net payoff as a function of q .
- (b) Derive the interior optimum and show that $q^* = \exp(1/\kappa)/(1 + \exp(1/\kappa))$.
- (c) Describe the limits as $\kappa \downarrow 0$ and $\kappa \uparrow \infty$.

Chapter 5

Choice under Uncertainty

5.1 Primitives

Let $Z \subseteq \mathbb{R}$ be a finite set of monetary prizes. A lottery is a probability mass function $p : Z \rightarrow [0, 1]$ satisfying $\sum_{z \in Z} p(z) = 1$. Write $\Delta(Z)$ for the set of lotteries and δ_z for the degenerate lottery assigning probability one to z . Mixtures are defined pointwise by

$$[\alpha p + (1 - \alpha)q](z) = \alpha p(z) + (1 - \alpha)q(z), \quad \alpha \in [0, 1].$$

For a Bernoulli utility $u : Z \rightarrow \mathbb{R}$, expected utility is $U(p) = \sum_{z \in Z} u(z)p(z)$.

5.2 Stochastic dominance on a finite prize set

Definition 5.1. First-order stochastic dominance. For lotteries $p, q \in \Delta(Z)$ with cumulative distribution functions $F_p(t) = \sum_{z \leq t} p(z)$ and $F_q(t) = \sum_{z \leq t} q(z)$, write $p \succeq^{FOSD} q$ if $F_p(t) \leq F_q(t)$ for every $t \in \mathbb{R}$. The relation is generally incomplete.

Theorem 5.2. If $p \succeq^{FOSD} q$ and u is increasing, then $U(p) \geq U(q)$ for expected utility U . Strict inequality follows when the dominance is strict on a set that receives positive probability and u is strictly increasing there.

5.3 Model and expected utility

We observe a choice correspondence on a family of subsets of $\Delta(Z)$ and represent the decision maker's preference by a relation $\succeq$ on lotteries.

Definition 5.3. Expected utility representation. The preference relation has an expected utility representation if there is a Bernoulli utility $u : Z \rightarrow \mathbb{R}$ such that $p \succeq q \iff \sum_{z \in Z} u(z)p(z) \geq \sum_{z \in Z} u(z)q(z)$.

5.4 Independence Axiom

We are assuming $\succeq$ are rational.

Axiom 5.4. (*Independence Axiom*) For all $p, q, r \in \Delta(Z)$ and $\alpha \in [0, 1]$, $p \succeq q$ implies $\alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r$.

The other axiom that we need is more technical

Axiom 5.5. (*Mixture continuity*) For all $p, q, r \in \Delta(Z)$, if $p \succ q \succ r$, then the set $\{\alpha \in [0, 1] | \alpha p + (1 - \alpha)r \succ q\}$ and $\{\alpha \in [0, 1] | q \succ \alpha p + (1 - \alpha)r\}$ are open in $[0, 1]$.

5.5 Representation Theorem

Theorem 5.6. The rational preference relation $\succeq$ satisfies the Independence Axiom and Mixture continuity if and only if there exists a (Bernoulli) utility function $u : Z \rightarrow \mathbb{R}$ such that: $p \succeq q$ if and only if $\sum_{z \in Z} u(z)p(z) \geq \sum_{z \in Z} u(z)q(z)$.

First we prove that there is an affine representation of $\succeq$.

Lemma 5.7. If the preference relation $\succeq$ satisfies the Independence Axiom and mixture continuity then there is an affine function $U : \Delta(Z) \rightarrow \mathbb{R}$ that represents $\succeq$.

Proof. Choose best and worst degenerate lotteries $p^+ = \delta_{z^+}$ and $p^- = \delta_{z^-}$ among the finitely many prizes. Repeated use of independence shows $p^+ \succeq p \succeq p^-$ for every lottery p . If $p^+ \sim p^-$, all lotteries are indifferent and a constant affine representation works. Otherwise, independence implies

$$\beta p^+ + (1 - \beta)p^- \succ \alpha p^+ + (1 - \alpha)p^- \iff \beta > \alpha.$$

For a fixed p , mixture continuity makes the two sets of mixture weights giving an outcome weakly above or weakly below p closed. They cover the connected interval $[0, 1]$ and contain opposite endpoints, so they intersect. Strict monotonicity along the best--worst segment gives a unique α_p with

$$p \sim \alpha_p p^+ + (1 - \alpha_p)p^-.$$

Thus $U(p) = \alpha_p$ represents $\succeq$. Finally, independence permits substitution of the equivalent best--worst mixtures for p and q , yielding

$$U(\beta p + (1 - \beta)q) = \beta U(p) + (1 - \beta)U(q).$$

Hence U is affine on the lottery simplex.

□

The second step is to show that an affine representation on the lottery simplex has the expected-utility form.

Lemma 5.8. *Let $U : \Delta(Z) \rightarrow \mathbb{R}$. Then U is affine if and only if there exists a $u : Z \rightarrow \mathbb{R}$ such that $U(p) = \sum_{z \in Z} u(z)p(z)$.*

For finite Z , no functional-analytic representation theorem is needed. Define $u(z) = U(\delta_z)$. Since every lottery has the finite decomposition $p = \sum_{z \in Z} p(z)\delta_z$, affinity of U gives $U(p) = \sum_{z \in Z} u(z)p(z)$. The converse follows immediately from distributivity.

Example 5.9. Let $Z = \{0, 10\}$, and let $p \in \Delta(Z)$ satisfy $p(0) = p(10) = \frac{1}{2}$. Then $U(p) = \frac{1}{2}u(10) + \frac{1}{2}u(0)$. Normalize the Bernoulli index by $u(0) = 0$ and $u(10) = 1$. The normalized expected utility is therefore $U(p) = 1/2$. Any positive affine transformation of this index represents the same preference.

5.6 Recoverability

Theorem 5.10. *Let $U : \Delta(Z) \rightarrow \mathbb{R}$ and $U' : \Delta(Z) \rightarrow \mathbb{R}$ be affine maps that represent $\succeq$ on $\Delta(Z)$, then there exist real numbers $\alpha > 0$ and b such that $U'(p) = \alpha U(p) + b$ for all $p \in \Delta(Z)$. If u, v are Bernoulli utilities such that $u \cdot p$ and $v \cdot p$ represent $\succeq$ then there are real numbers $\alpha > 0$ and b such that $v \cdot p = \alpha u \cdot p + b$.*

Think about this. How it differs from the recoverability result of the common rational preference $\succeq$.

5.7 Attitudes towards Risk

Assume that Bernoulli utility is defined on the convex hull $\text{co}(Z)$ of the finite monetary-prize set. For a lottery p , its expected monetary value is $\bar{z}(p) = \sum_{z \in Z} zp(z)$ and the corresponding certain outcome is $\delta_{\bar{z}(p)}$.

Definition 5.11. Risk aversion. A decision maker is risk averse if $\delta_{\bar{z}(p)} \succeq p$ for every lottery p . Under expected utility, risk aversion is equivalent to concavity of the Bernoulli utility:

$$\sum_{z \in Z} p(z)u(z) \leq u\left(\sum_{z \in Z} p(z)z\right).$$

Certainty equivalent and risk premium. If u is strictly increasing and its range contains $U(p)$, the certainty equivalent is the prize $\text{CE}(p, u) = u^{-1}(U(p))$ and the (monetary) risk premium is $\rho(p, u) = \bar{z}(p) - \text{CE}(p, u)$. A risk-averse expected-utility decision maker has $\rho(p, u) \geq 0$.

Probability premium. For prizes $x - \varepsilon, x, x + \varepsilon$ in the certainty set, $\pi(x, \varepsilon, u)$ is defined by

$$u(x) = \left(\frac{1}{2} + \pi\right)u(x + \varepsilon) + \left(\frac{1}{2} - \pi\right)u(x - \varepsilon).$$

It is the change in probability needed to make the symmetric two-point lottery equivalent to the sure prize.

Proposition 5.12. For a twice differentiable, strictly increasing Bernoulli utility, the local risk premium for a symmetric small spread satisfies

$$\rho\left(\frac{1}{2}\delta_{x-\varepsilon} + \frac{1}{2}\delta_{x+\varepsilon}, u\right) = -\frac{u''(x)}{2u'(x)}\varepsilon^2 + o(\varepsilon^2).$$

Thus local concavity corresponds to a nonnegative local risk premium.

5.8 Expected Utility for Infinite Sets

Let Z be a compact metric prize space and let Σ be its Borel σ -algebra. Recall that a collection of subsets is a σ -algebra when

- $\emptyset \in \Sigma$
- if $E \in \Sigma$ then $E^c \in \Sigma$
- If $\{A_n\}$ is a countable collection of sets such that $A_n \in \Sigma$ then $\cup_n A_n \in \Sigma$.

We say that (Z, Σ) is a measurable space whenever Σ is a σ -algebra.

Definition 5.13. We say that $p : \Sigma \rightarrow [0, 1]$ is a probability measure if:

- $p(Z) = 1$

- if $\{A_n\}$ is a countable collection of pairwise disjoint sets such that $A_n \in \Sigma$, then $p(\cup_n A_n) = \sum_n p(A_n)$.

We call the collection of probability measures $\Delta(Z, \Sigma)$ for short $\Delta(Z)$ when Σ is fixed.

$\alpha p + (1 - \alpha)q \in \Delta(Z)$ is defined as $(\alpha p + (1 - \alpha)q)(E) = \alpha p(E) + (1 - \alpha)q(E)$.

Equip this space with weak convergence: $p^n \Rightarrow p$ if $\int f dp^n \rightarrow \int f dp$ for continuous and bounded functions f .

Axiom 5.14. (*Weak continuity*) For any $p \in \Delta(Z)$ the sets $\{q \in \Delta(Z) : q \succ p\}$ and $\{q \in \Delta(Z) : p \succ q\}$ are open in the weak topology.

Theorem 5.15. Let Z be a compact metric prize space and $\Delta(Z)$ the Borel probability measures with the weak topology. A rational preference relation $\succeq$ on $\Delta(Z)$ satisfies the independence axiom and weak continuity if and only if there exists a $u : Z \rightarrow \mathbb{R}$ bounded and continuous such that $U(p) = \int u(z)dp(z)$ represents $\succeq$.

5.9 Application: Demand for a Risky Asset

An asset is a divisible claim to a financial return in the future. Assume that u is continuously differentiable, strictly increasing, and concave on an interval containing every feasible terminal-wealth realization. Assume also that the return and marginal-utility expressions below are integrable.

Let α, β denote the amounts of money invested in a risky and safe asset respectively. The risky asset pays a random return of Z dollars per dollar invested and the safe asset pays 1 dollar with certainty per dollar invested. The random return has a distribution $F(z)$ (CDF). We assume that $\int z dF(z) > 1$, or that the risky asset expected return exceeds the return of the safe asset. Then the total return is $\alpha Z + \beta$. Feasibility requires $\alpha + \beta = w$.

$$\max_{\alpha, \beta \geq 0} \int u(\alpha z + \beta) dF(z)$$

$$\text{subject to } \alpha + \beta = w.$$

Equivalently we want to solve.

$$\max_{0 \leq \alpha \leq w} \int u(w + \alpha(z - 1)) dF(z).$$

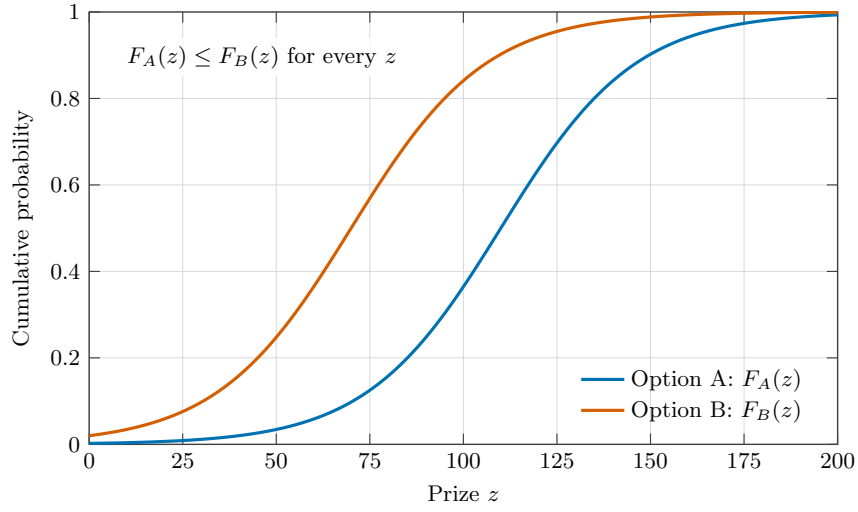


Figure 5.1: First-order stochastic dominance. Option A is a rightward location shift of Option B, so $F_A(z) \leq F_B(z)$ for every payoff z .

The first-order conditions are:

$$\phi(\alpha^*) = \int u'(w + \alpha^*[z - 1])[z - 1]dF(z) \begin{cases} \leq 0 & \text{if } \alpha^* < w \\ \geq 0 & \text{if } \alpha^* > 0. \end{cases}$$

Note that $\int z dF(z) > 1$ implies that $\phi(0) > 0$.

Because $u'(w) > 0$.

And $\phi(0) = \int u'(w)[z - 1]dF(z)$.

So $\alpha^* = 0$ cannot satisfy the first order conditions.

If a risk is actuarially favorable, then a risk-averse expected-utility maximizer will accept at least a sufficiently small amount under the stated differentiability and interior-wealth assumptions.

5.10 Stochastic dominance for general distributions

This section extends the finite-prize result proved earlier in the chapter to probability distributions with integrable utility.

Theorem 5.16. *First-order stochastic dominance. If $F \succeq^{FOSD} G$, then $\int u dF \geq \int u dG$ for every increasing Bernoulli utility u for which the integrals exist. Strict dominance together with strict monotonicity yields a strict inequality under the usual support conditions.*

$F \succeq^{FOSD} G$ FOSD means $F(x) \leq G(x)$ for all $x \in Z$ with strict inequality for some $x \in Z$.

5.11 The Allais Paradox

Z is finite in fact $N = 3$,

First Prize	Second Prize	Third Prize
2.5 millions	0.5 millions	0

Two choice sets:

The first,

$$L_1 = (0, 1, 0) \text{ and } L'_1 = (0.10, 0.89, 0.01).$$

The second,

$$L_2 = (0, 0.11, 0.89) \text{ and } L'_2 = (0.10, 0, 0.90).$$

Some individuals choose $L_1 \succ L'_1$ and $L'_2 \succ L_2$.

This is inconsistent with expected utility.

If $L_1 \succ L'_1$ then

$$u_{0.5} > (0.10)u_{2.5} + (0.89)u_{0.5} + (0.01)u_0.$$

Adding, $(0.89)u_0 - (0.89)u_{0.5}$ to both sides we get,

$$(0.11)u_{0.5} + (0.89)u_0 > (0.1)u_{2.5} + (0.9)u_0,$$

and thus $L_2 \succ L'_2$.

5.12 Revealed Preference of Expected Utility under Concavity

Let S be a finite set of states with fixed probabilities $\pi_s > 0$, and let a contingent consumption plan be $x = (x_s)_{s \in S} \in \mathbb{R}_{++}^S$. At prices $p \in \mathbb{R}_{++}^S$ and wealth $w > 0$, the consumer solves $\max_x \sum_{s \in S} \pi_s u(x_s)$ subject to $\sum_{s \in S} p_s x_s \leq w$, where u is continuous, strictly increasing and concave, with positive marginal utility on the observed consumption range whenever derivatives are invoked.

Proposition 5.17. *Interior first-order conditions. If u is differentiable and an optimum is interior, then there is $\lambda > 0$ such that $\pi_s u'(x_s) = \lambda p_s$ for every $s \in S$. Concavity makes these conditions sufficient for a global optimum; boundary observations require the corresponding KKT inequalities.*

Definition 5.18. Risk-averse expected-utility rationalization. A finite dataset $\mathcal{O}^T = \{(p^t, x^t)\}_{t=1}^T$ is rationalized if there is one continuous, strictly increasing and concave Bernoulli utility with positive marginal utility on the observed range u such that each x^t maximizes $\sum_s \pi_s u(x_s)$ on the budget set $\{x : p^t \cdot x \leq p^t \cdot x^t\}$.

Theorem 5.19. *Finite-data test.* See [Polisson et al., 2020]. Assume all observations are interior and $\pi_s > 0$. The dataset has a risk-averse expected-utility rationalization if and only if there are numbers $\beta_s^t > 0$ and $\lambda^t > 0$ that satisfy the price equations

$$\pi_s \beta_s^t = \lambda^t p_s^t \quad \text{for every } s, t,$$

and the monotonicity inequalities

$$x_s^t > x_{s'}^{t'} \implies \beta_s^t \leq \beta_{s'}^{t'}$$

for every pair of observed state-contingent quantities. The monotonicity condition is the finite signature of a single decreasing marginal-utility function; the price equations are the common first-order conditions.

The theorem is an Afriat-style interpolation result for one Bernoulli utility across states and observations. It is stated for interior data to keep the notation transparent; zero quantities are handled by replacing equalities with KKT inequalities.

5.13 Experimental Evidence

Polisson and Quah develop a revealed preferences test for expected utility. They apply their framework to the famous Choi, Fisman, Gale and Kariv (2007) dataset and find the following passing rates.

- GARP is the benchmark revealed-preference test.
- RDEU denotes rank-dependent expected utility.
- EU denotes expected utility.
- EU* imposes concavity of the Bernoulli utility function in addition to expected utility.

The “drop ” column reports how many observations the test permits the researcher to discard.

Observations dropped	GARP	RDEU	EU	EU*
0	12/47 (26%)	2/47 (4%)	2/47 (4%)	0/47 (0%)
1	14/47 (30%)	7/47 (15%)	5/47 (11%)	0/47 (0%)
2	27/47 (57%)	9/47 (19%)	7/47 (15%)	2/47 (4%)
3	32/47 (68%)	12/47 (26%)	10/47 (21%)	2/47 (4%)

Figure 5.2: Share of 47 subjects passing each revealed-preference test after dropping up to three observations [Polisson and Quah, 2013].

5.14 Measuring Risk Aversion in the Lab

The Holt and Laury multiple-price-list task classifies each subject by the total number of safe (Option A) choices across ten decisions.

Assume constant relative risk aversion (CRRA): $u(x) = x^{1-r}/(1-r)$. At $r = 1$, use $u(x) = \log x$. A monotone pattern with k initial Option A choices followed by Option B implies an interval for r .

Number of safe choices	CRRA interval for $U(x) = x^{1-r}/(1-r)$	Proportion of choice patterns	
		Low real ($\times 1$) ^a	Real ($\times 20$)
0–1	$r < -0.95$	.01	.01
2	$-0.95 < r < -0.49$	.01	.01
3	$-0.49 < r < -0.15$	.06	.04
4	$-0.15 < r < 0.15$	.26	.13
5	$0.15 < r < 0.41$	.26	.19
6	$0.41 < r < 0.68$	.23	.23
7	$0.68 < r < 0.97$	.13	.22
8	$0.97 < r < 1.37$	.03	.11
9–10	$1.37 < r$	.01	.06

^aAverage over the first and second low-real decisions.

Figure 5.3: Distribution of the number of safe choices in the Holt–Laury task. The final two columns give proportions of observed choice patterns classified by total number of Option A choices under low real ($\times 1$) and real $\times 20$ payoffs. The low-real column averages the first and second low-real decisions [Holt and Laury, 2002].

Individuals choose between A and B in each of the ten rows.

Each option has two monetary outcomes, A_1, A_2 and B_1, B_2 ; their common high-outcome probability rises by one tenth from one row to the next.

The expected payoff of A exceeds that of B in rows 1--4; the inequality reverses in rows 5--10.

- A risk-neutral expected-utility maximizer chooses A in rows 1, ..., 4 and B in rows 5, ..., 10.
- Option B has higher variance in each nondegenerate row (rows 1--9).
- Under expected utility and CRRA, five or more initial safe choices imply $r > 0$, and a later single switch to B corresponds to a larger risk-aversion parameter. Choosing A in row 10 violates monotonicity because both options are then certain and $3.85 > 2.00$.

One row is selected at random for payment after all choices are recorded. This random-lottery incentive system makes every row potentially payoff relevant; interpreting the switching row as a risk-aversion measure additionally requires expected utility, monotonicity, and the stated payoff probabilities.

5.15 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 5.1 [Core]. Affine expected-utility representations.

Let $Z = \{z_1, \dots, z_n\}$ and suppose $U(p) = \sum_j p_j u(z_j)$ represents a nontrivial preference on $\Delta(Z)$.

- Show directly that U is affine in lotteries.
- Prove that $U' = aU + b$ with $a > 0$ represents the same preference.
- Show that if two expected-utility representations agree on preferences, their Bernoulli indices differ by a positive affine transformation.

Problem 5.2 [Core]. First-order stochastic dominance.

Lotteries p and q on prizes $\{0, 1, 2\}$ have probabilities $p = (0.2, 0.3, 0.5)$ and $q = (0.3, 0.4, 0.3)$.

- Compare their cumulative distribution functions and determine which lottery first-order stochastically dominates the other.
- Prove the expected-utility ranking for every increasing Bernoulli utility by discrete summation by parts.
- Explain why strict dominance need not imply a strict preference for a weakly increasing utility.

Problem 5.3 [Proof]. The Arrow-Pratt approximation.

An agent with twice continuously differentiable utility u and wealth w faces a zero-mean risk ε with variance σ^2 . Define the risk premium π by $u(w - \pi) = \mathbb{E}[u(w + \varepsilon)]$.

- (a) Use second-order Taylor expansions to derive $\pi \approx \frac{1}{2}A(w)\sigma^2$, where $A(w) = -u''(w)/u'(w)$.
- (b) State the regularity and small-risk qualifications needed for this approximation.
- (c) Explain why the formula is invariant to positive affine transformations of u .

Problem 5.4 [Applied]. CARA-normal portfolio demand.

Terminal wealth is $W = wR_f + \theta(R - R_f)$, where $R \sim N(\mu, \sigma^2)$, R_f is risk free, $\theta \in \mathbb{R}$ (borrowing and short selling are unrestricted), and utility is $u(W) = -\exp(-aW)$ with $a > 0$.

- (a) Show that maximizing expected utility is equivalent to maximizing a certainty equivalent.
- (b) Derive optimal risky investment θ^* .
- (c) Give comparative statics with respect to $\mu - R_f$, a , and σ^2 .

Part III

Aggregation and Random Choice

Chapter 6

Aggregate Demand

6.1 Three questions about aggregation

- When does aggregate demand depend on prices and total wealth rather than on the entire wealth distribution?
- When does aggregate demand satisfy a demand restriction such as WARP?
- When can aggregate demand be given a welfare interpretation?

Let consumers be indexed by $i = 1, \dots, I$. Consumer i has Marshallian demand $x_i(p, w_i)$ at common prices $p \in \mathbb{R}_{++}^L$ and wealth $w_i > 0$. Aggregate demand is

$$X(p, \mathbf{w}) = \sum_{i=1}^I x_i(p, w_i),$$

where $\mathbf{w} = (w_1, \dots, w_I)$. Total wealth is $W = \sum_i w_i$. Without additional structure, two wealth distributions with the same W can generate different aggregate bundles.

6.2 Gorman aggregation

Definition 6.1. Gorman form. Consumer i has Gorman-form indirect utility if

$$v_i(p, w_i) = a_i(p) + b(p)w_i,$$

for a common function $b(p) > 0$ and consumer-specific functions $a_i(p)$.

Proposition 6.2. *Gorman aggregation. See [Gorman, 1953, Mas-Colell et al., 1995]. Suppose each v_i has the Gorman form and is differentiable. Roy's identity gives*

$$x_i(p, w_i) = -\frac{\nabla_p a_i(p) + w_i \nabla_p b(p)}{b(p)}.$$

Therefore aggregate demand takes the form

$$X(p, \mathbf{w}) = A(p) + B(p)W,$$

where $A(p) = -\sum_i \nabla_p a_i(p)/b(p)$ and $B(p) = -\nabla_p b(p)/b(p)$. Aggregate demand consequently depends on (p, W) and not on the distribution of wealth.

Under the usual differentiability and integrability conditions, a common wealth derivative $D_{w_i} x_i(p, w_i) = B(p)$ is the local signature of Gorman aggregation. The converse is a local statement and should not be read as a global representation without the regularity assumptions.

6.3 Aggregate demand and WARP

Even if every individual demand satisfies WARP, aggregate demand need not do so. Aggregate WARP compares two price--total-wealth pairs while holding fixed the rule that assigns total wealth across consumers; the corresponding compensation need not leave each consumer's wealth unchanged.

Definition 6.3. *Aggregate WARP. A single-valued aggregate demand $X(p, W)$ satisfies WARP if, for distinct observations (p, W) and (p', W') , $p \cdot X(p', W') \leq W$ implies $p' \cdot X(p, W) > W'$ whenever $X(p, W) \neq X(p', W')$.*

Proposition 6.4. *Sufficient condition for aggregate WARP. Suppose wealth shares are fixed and strictly positive, $w_i = \alpha_i W$ with $\alpha_i > 0$ and $\sum_i \alpha_i = 1$. Suppose every individual demand is homogeneous of degree zero and satisfies the weak uncompensated law of demand on the normalized-price domain. With $q = p/W$, homogeneity gives $x_i(p, \alpha_i W) = x_i(q, \alpha_i)$. Then $X(p, W) = \sum_i x_i(p, \alpha_i W)$ satisfies the weak uncompensated law of demand. With Walras' law and the usual strictness condition, aggregate WARP follows.*

The proof first normalizes each price--total-wealth pair to unit total wealth and then sums the individual inequalities at the common normalized wealth shares α_i . If the shares change with prices or if wealth is reallocated after compensation, this argument no longer applies.

6.4 Representative consumers and welfare

Definition 6.5. Positive representative consumer. A preference relation $\succeq^{rep}$ is a positive representative consumer for X if its Walrasian demand equals $X(p, W)$ for every price--wealth pair in the domain. This is a statement about demand representation, not about welfare.

Normative representative consumer. Fix a social welfare function $\mathcal{W} : \mathbb{R}^I \rightarrow \mathbb{R}$ and individual indirect utilities v_i . A nonnegative wealth allocation $\mathbf{w}(p, W)$ is welfare-optimal if it solves

$$\max_{\mathbf{w} \in \mathbb{R}_+^I} \mathcal{W}(v_1(p, w_1), \dots, v_I(p, w_I)) \quad \text{subject to} \quad \sum_i w_i \leq W.$$

. A positive representative consumer is normative relative to $\mathcal{W}$ only when the wealth rule used to form X is welfare-optimal at every (p, W) .

Gorman aggregation can deliver a positive representative consumer, but it does not select a social welfare function or justify interpersonal utility comparisons. Normative claims require the welfare criterion and the wealth-allocation rule to be stated explicitly.

6.5 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 6.1 [Core]. Gorman aggregation.

Suppose consumer i has indirect utility $v_i(p, w_i) = a_i(p) + b(p)w_i$, with the same positive function b for every consumer.

- Apply Roy's identity to derive individual demand.
- Show that aggregate demand depends on the wealth distribution only through total wealth $W = \sum_i w_i$.
- Construct a representative indirect utility that generates aggregate demand.

Problem 6.2 [Core]. Aggregation under common homothetic preferences.

All consumers have the same homothetic preferences, strictly positive wealth $w_i > 0$, and a single-valued demand. Normalize the unit-wealth demand as $x(p, 1)$.

- Show that $x_i(p, w_i) = w_i x(p, 1)$.
- Derive aggregate demand and identify a representative consumer.
- Explain why common homotheticity is sufficient but not necessary for exact aggregation.

Problem 6.3 [Proof]. Why uncompensated aggregate price derivatives need not be symmetric.

Let $x^A(p) = \sum_i x_i(p, w_i(p))$, where individual wealth may vary with prices through endowments or profit shares.

- (a) Differentiate x^A with respect to prices.
- (b) Separate compensated substitution terms from wealth-revaluation terms.
- (c) Explain why symmetry and negative semidefiniteness of each individual Slutsky matrix do not generally imply the same properties for the uncompensated derivative $D_p x^A$.

Chapter 7

Discrete Choice and Random Utility

7.1 Choice from a finite menu

There are alternatives $j \in \{0, 1, \dots, J\}$, where $j = 0$ is the outside option. Consumer i chooses one alternative from the menu after comparing conditional indirect utilities:

$$U_{ij}^* = V_j(x_j, p_j, y_i) + \varepsilon_{ij}.$$

Here x_j denotes observed attributes, V_j is mean utility, and ε_{ij} is an unobserved preference shock.

The choice probability is

$$P_{ij} = \Pr\{U_{ij}^* \geq U_{ik}^* \text{ for every } k\},$$

with strict inequalities almost surely when the shock distribution is continuous.

If the outside good has price p_0 and quantity z , a budget identity $p_j + p_0 z = y_i$ yields $z = (y_i - p_j)/p_0$ for a purchased alternative. This reduction is valid only when the implied outside-good quantity is feasible.

7.2 Multinomial logit and IIA

Suppose the shocks are independent and identically distributed Type-I extreme value with scale $\mu > 0$. Then the multinomial-logit probabilities are

$$P_{ij} = \frac{\exp(V_j/\mu)}{\sum_{k=0}^J \exp(V_k/\mu)}.$$

Proposition 7.1. *Independence of irrelevant alternatives. Under multinomial logit, for any two alternatives j, k that remain available, $\frac{P_{ij}}{P_{ik}} = \exp((V_j - V_k)/\mu)$ is independent of the other alternatives in the menu.*

Example 7.2. Red-bus/blue-bus. If walking and a red bus initially have equal probabilities, adding a blue bus that is nearly identical to the red bus should split the bus share while leaving walking's share close to its original value. Multinomial logit instead preserves the walking-to-red-bus odds ratio, illustrating the empirical content and limitation of IIA.

The multinomial probit model replaces the extreme-value shocks with a jointly normal vector. Its pairwise differences are generally correlated, so it does not impose IIA.

7.3 Random utility

Definition 7.3. Random-utility representation. Let A be a finite menu and let $\mathbf{U} : \Omega \rightarrow \mathbb{R}^A$ be a measurable random utility vector. The induced choice probabilities are $\rho_A(a) = \mathbb{P}\{\mathbf{U}(a) > \mathbf{U}(b) \text{ for every } b \in A \setminus \{a\}\}$. Assume either that ties occur with probability zero or that a measurable tie-breaking rule is specified; under the zero-tie convention the displayed strict-max probabilities sum to one.

When A is finite and choice probabilities are observed on every nonempty menu, the deterministic types are the strict linear orders on A . Stacking the probabilities gives $\rho = M\pi$, where $\pi \in \Delta(R)$ is a distribution over those orders and $M_{(a,B),r} = \mathbb{I}\{a \text{ is } r\text{-best in } B\}$. The finite feasibility condition $\rho = M\pi$ with $\pi \geq 0$ and $\mathbf{1} \cdot \pi = 1$ is equivalent to the existence of such a distribution.

7.4 Random expected utility

Definition 7.4. Random expected utility. For a finite prize set Z and lotteries $p \in \Delta(Z)$, a random Bernoulli utility $\mathbf{u} : \Omega \rightarrow \mathbb{R}^Z$ induces $\mathbf{U}_\omega(p) = \sum_{z \in Z} \mathbf{u}_\omega(z)p(z)$. Choice probabilities

are obtained by maximizing this random expected utility on each menu of lotteries.

The representation imposes linearity in mixtures and regularity under menu expansion. Extremeness and weak-star continuity are additional conditions in the Gul–Pesendorfer characterization; see [Gul and Pesendorfer, 2006].

7.5 Attributes and consumer surplus

With observed attributes $x = (x_a)_{a \in A}$, a common specification is

$$U(a, x) = v(a, x_a) + \varepsilon_a,$$

where the distribution of $\varepsilon = (\varepsilon_a)_{a \in A}$ is independent of x . The linear-index model uses $v(a, x_a) = x_a^\top \beta$. Independence is a maintained exogeneity assumption, not a consequence of the random-utility definition.

Definition 7.5. Expected maximum utility. The expected value of a menu is the function

$$V(x) = \mathbb{E} \left[\max_{a \in A} \{v(a, x_a) + \varepsilon_a\} \right].$$

If the deterministic utility contains $-\alpha p_a$ with $\alpha > 0$ as the price coefficient, the publication-safe money-metric object is the change between two scenarios, $\Delta CS = [V(x^1) - V(x^0)]/\alpha$. The level $V(x)/\alpha$ depends on the additive utility normalization. If shocks are merely decision errors rather than preference shocks, this expression is not a welfare measure.

Proposition 7.6. *Williams--Daly--Zachary envelope result. See [McFadden, 1981]. If the shock distribution has finite first moments and ties occur with probability zero, then V is convex and differentiable in the vector of mean utilities, and $\frac{\partial V(x)}{\partial v(a)} = \rho(a, x)$. If choice probabilities are differentiable in mean utilities (equivalently, if V is twice differentiable at the point), then the Jacobian of choice probabilities is the Hessian of V , hence symmetric and positive semidefinite. Translation invariance implies that the rows and columns sum to zero.*

The envelope identity is the basis for likelihood and welfare calculations in discrete-choice models. Additional sign restrictions such as monotone substitution patterns require additional assumptions on the shock distribution and utility index; they do not follow from random utility alone.

7.6 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 7.1 [Core]. Logit odds and IIA.

Choice probabilities satisfy $P(j | A) = \exp(V_j/\mu) / \sum_{k \in A} \exp(V_k/\mu)$, where $\mu > 0$.

- (a) Derive the odds ratio $P(j | A)/P(k | A)$.
- (b) Prove the independence-of-irrelevant-alternatives property.
- (c) Explain precisely which part of IIA is an empirical restriction and which part is an algebraic implication of the model.

Problem 7.2 [Applied]. The red-bus blue-bus problem.

Walking and a red bus initially have systematic utilities normalized so that each has probability one half under logit. A blue bus with the same systematic utility as the red bus is added.

- (a) Compute the new logit probabilities.
- (b) Compare the total bus share before and after entry.
- (c) Explain why a model with highly correlated bus-specific shocks would predict a different substitution pattern.

Problem 7.3 [Core]. A finite random-utility mixture.

There are alternatives a, b, c . A population consists of three strict preference types: $a \succ b \succ c$ with mass 0.5, $c \succ a \succ b$ with mass 0.2, and $b \succ a \succ c$ with mass 0.3.

- (a) Compute choice probabilities from menus $\{a, b\}$ and $\{a, b, c\}$.
- (b) Let the columns of M correspond to all six strict orders on $\{a, b, c\}$, with zero masses allowed. Write these probabilities in the form $\rho = M\pi$.
- (c) Explain why nonnegativity of π and $\mathbf{1}^\top \pi = 1$ are the central finite random-utility feasibility restrictions.

Part IV

Production and Equilibrium

Chapter 8

Theory of the Firm

This chapter treats technology as a feasible set of net-output vectors. Profit maximization and cost minimization are dual ways of valuing that set at prices; aggregation and efficiency follow from the same support-function geometry.

8.1 Production sets

There are $L \geq 1$ commodities. A production plan $y \in \mathbb{R}^L$ records net outputs: $y_\ell > 0$ is an output and $y_\ell < 0$ is an input. The production set $Y \subseteq \mathbb{R}^L$ contains exactly the technologically feasible plans.

The benchmark assumptions are: Y is nonempty and closed; $0 \in Y$ (inaction); and free disposal, $y \in Y$ and $y' \leq y$ imply $y' \in Y$. A no-free-lunch condition rules out $y \in Y \cap \mathbb{R}_+^L$ other than 0. Convexity of Y is the convex-technology assumption. Together with $0 \in Y$, it guarantees feasibility of proportional contractions, but it does not impose decreasing returns to scale for expansions.

If a differentiable transformation function satisfies $Y = \{y : F(y) \leq 0\}$ and $F(\bar{y}) = 0$ with $F_k(\bar{y}) \neq 0$, the implicit-function theorem gives the local slope of the transformation frontier:

$$\frac{dy_k}{dy_\ell} = -\frac{F_\ell(\bar{y})}{F_k(\bar{y})}.$$

The minus sign is essential: increasing one net output requires a compensating reduction in the other when the frontier is written as $F(y) = 0$.

8.2 Production functions and input substitution

With M outputs and $L - M$ inputs, write outputs as $q \in \mathbb{R}_+^M$ and input use as $z \in \mathbb{R}_+^{L-M}$. In the single-output case a production function f describes free disposal through $q \leq f(z)$. If f is differentiable, the marginal rate of technical substitution between inputs ℓ and k is

$$\text{MRTS}_{\ell k}(z) = \frac{f_\ell(z)}{f_k(z)},$$

the amount of input k needed to offset a marginal reduction in input ℓ while holding output fixed. Concavity of f makes the input requirement set $\{z : f(z) \geq q\}$ convex.

8.3 Profit maximization

Definition 8.1. For a price vector $p \in \mathbb{R}^L$, the profit function is

$$\pi(p) = \sup_{y \in Y} p \cdot y,$$

and the supply correspondence is $y(p) = \arg \max_{y \in Y} p \cdot y$ whenever the supremum is finite and attained. If Y is closed and unbounded, finiteness and attainment require separate assumptions.

Proposition 8.2. *Profit-function properties. See [Hotelling, 1932, Mas-Colell et al., 1995]. The function π is convex and homogeneous of degree one in prices. The supply correspondence is homogeneous of degree zero; if Y is convex, it is convex-valued. At a differentiability point, Hotelling's lemma gives $\nabla \pi(p) = y(p)$. If π is twice differentiable, $D_p y(p) = D_p^2 \pi(p)$ is symmetric positive semidefinite and $D_p y(p)p = 0$.*

The non-differentiable law of supply is the monotonicity inequality

$$(p - p') \cdot (y - y') \geq 0$$

for $y \in y(p)$ and $y' \in y(p')$. It follows by adding the two profit-maximization inequalities.

For a single-output technology, $q = f(z)$, profit maximization is $\max_{z \geq 0} \{pf(z) - w \cdot z\}$. At a differentiable interior optimum, $p \nabla f(z^*) = w$; with nonnegative inputs the KKT conditions are

$$pf_\ell(z^*) \leq w_\ell, \quad z_\ell^*(w_\ell - pf_\ell(z^*)) = 0.$$

8.4 Cost minimization

Definition 8.3. Given input prices $w \gg 0$ and a target $q > 0$, the cost function is

$$c(w, q) = \min_{z \geq 0} \{w \cdot z : f(z) \geq q\}.$$

Assume the minimum is finite and attained. The conditional factor-demand correspondence is the argmin set $z(w, q)$. At a differentiable optimum satisfying a constraint qualification, the complete KKT system is

$$\begin{aligned} z^* &\geq 0, \quad f(z^*) \geq q, \quad \lambda \geq 0, \\ w_\ell - \lambda f_\ell(z^*) &\geq 0, \quad z_\ell^*(w_\ell - \lambda f_\ell(z^*)) = 0, \\ \lambda(f(z^*) - q) &= 0. \end{aligned}$$

Proposition 8.4. *Cost-function properties. See [Shephard, 1953, Mas-Colell et al., 1995]. The cost function is homogeneous of degree one and concave in input prices, nondecreasing in the output target, and—when f is concave—convex in q . If the cost function is differentiable, Shephard's lemma gives $\nabla_w c(w, q) = z(w, q)$. If, in addition, the output constraint binds, the relevant constraint qualification holds, and the value function is differentiable in q , then the envelope theorem gives $\partial_q c(w, q) = \lambda$. If twice differentiable, $D_w z = D_w^2 c$ is symmetric negative semidefinite and $D_w z w = 0$ by price homogeneity.*

The firm's output choice can be written as $\max_{q \geq 0} \{pq - c(w, q)\}$. If the cost function is differentiable, an interior optimum satisfies $p = \partial_q c(w, q)$; at $q = 0$ the corresponding one-sided inequality applies.

8.5 Aggregation and efficiency

For firms $j = 1, \dots, J$ with production sets Y_j , the aggregate technology is the Minkowski sum

$$Y = \sum_{j=1}^J Y_j = \left\{ \sum_{j=1}^J y_j : y_j \in Y_j \right\}.$$

Proposition 8.5. *If each firm's profit maximum is finite and attained, then*

$$\pi_Y(p) = \sum_{j=1}^J \pi_{Y_j}(p), \quad y_Y(p) = \sum_{j=1}^J y_j(p)$$

where the second equality is understood as equality of attainable aggregate supply sets. The first equality follows directly from separability of the maximization over the Minkowski sum.

Definition 8.6. A feasible production plan $y \in Y$ is efficient if there is no $y' \in Y$ with $y' \geq y$ and $y' \neq y$.

Proposition 8.7. *If $p \gg 0$ and $y \in y(p)$, then y is efficient. If Y is convex and closed and satisfies free disposal, every efficient plan is supported by some nonzero $p \geq 0$; this supporting-price statement is a separating-hyperplane theorem and may allow zero prices.*

8.6 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 8.1 [Core]. Cobb-Douglas production and cost.

A firm has production function $f(z) = z_1^\alpha z_2^\beta$ with $\alpha, \beta > 0$.

- (a) Classify returns to scale as a function of $\alpha + \beta$.
- (b) Derive conditional factor demands and the cost function for output $q > 0$ and input prices $w \gg 0$.
- (c) Verify that cost is homogeneous of degree one and concave in w .

Problem 8.2 [Proof]. Dual properties of the profit function.

Let $Y \subseteq \mathbb{R}^L$ be nonempty, closed, and convex. Assume that profit $\pi(p) = \sup_{y \in Y} p \cdot y$ is finite on a neighborhood of the price under study and that the supremum is attained.

- (a) Prove that π is convex and homogeneous of degree one in prices.
- (b) If supply is unique at p , use a subgradient argument to prove Hotelling's lemma and explain why the profit function is differentiable there.
- (c) State what changes when supply is set-valued.

Problem 8.3 [Core]. Constant returns and unbounded profit.

Assume Y is a cone and $0 \in Y$.

- (a) Prove that either $\pi(p) = 0$ or $\pi(p) = +\infty$.
- (b) Characterize the price vectors for which profit is zero.
- (c) Explain why a competitive constant-returns firm can operate at any scale only when the zero-profit condition holds.

Problem 8.4 [Proof]. Shephard's lemma and substitution.

Suppose the minimum in $c(w, q) = \min\{w \cdot z : f(z) \geq q, z \geq 0\}$ exists, conditional factor demand $z(w, q)$ is unique, and c is twice differentiable in w .

- (a) Prove Shephard's lemma.
- (b) Show that $D_w z = D_{ww}^2 c$ is symmetric and negative semidefinite.
- (c) Use homogeneity to show $D_w z(w, q)w = 0$.

Chapter 9

Competitive Markets and Partial Equilibrium

This chapter collects the competitive-equilibrium definitions used later and develops a transparent partial-equilibrium model with a numeraire. The partial-equilibrium model is deliberately specialized: its quasilinear preferences remove income effects for the good of interest.

9.1 Competitive equilibrium

There are I consumers, J firms, and L goods. Consumer i has consumption set $X_i \subseteq \mathbb{R}^L$, preference relation $\succeq_i$, and endowment $\omega_i \in \mathbb{R}^L$. Firm j has production set $Y_j \subseteq \mathbb{R}^L$, where positive coordinates are net outputs and negative coordinates are inputs. Let $\omega = \sum_i \omega_i$.

Definition 9.1. Feasibility. An allocation $(x_1, \dots, x_I; y_1, \dots, y_J)$ is feasible when $x_i \in X_i$, $y_j \in Y_j$, and

$$\sum_{i=1}^I x_i \leq \omega + \sum_{j=1}^J y_j.$$

Equality is imposed when all goods have strictly positive prices and free disposal is not used.

Competitive (Walrasian) equilibrium. A nonzero nonnegative price vector $p \geq 0$ and an allocation (x^*, y^*) form an equilibrium if every firm solves

$$y_j^* \in \arg \max_{y_j \in Y_j} p \cdot y_j$$

and every consumer weakly prefers their assignment to every affordable bundle:

$$x_i^* \succeq_i x_i \quad \text{for every } x_i \in X_i \text{ with } p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*.$$

Ownership shares satisfy $\theta_{ij} \geq 0$ and $\sum_i \theta_{ij} = 1$ for each firm. Markets clear exactly: $\sum_i x_i^* = \omega + \sum_j y_j^*$.

Proposition 9.2. *Price normalization and Walras' law. If (p, x^*, y^*) is a competitive equilibrium, then so is $(\lambda p, x^*, y^*)$ for every $\lambda > 0$. If consumers exhaust their budgets, summing the budget equalities and using market clearing gives*

$$p \cdot \left(\sum_i x_i^* - \omega - \sum_j y_j^* \right) = 0.$$

Therefore, if all but one market clear, the last market clears as well.

9.2 Partial equilibrium with a numeraire

There are two goods. Good x is the good of interest and m is a numeraire with price one. Consumer i has

$$u_i(m_i, x_i) = m_i + \phi_i(x_i), \quad X_i = \mathbb{R} \times \mathbb{R}_+,$$

where ϕ_i is increasing, strictly concave, twice continuously differentiable, and bounded above.

Let $p > 0$ be the price of x .

Firm j produces $q_j \geq 0$ units of x at numeraire cost $c_j(q_j)$. Its net-output set is

$$Y_j = \{(-z_j, q_j) : q_j \geq 0, z_j \geq c_j(q_j)\}.$$

Assume c_j is increasing, strictly convex, twice continuously differentiable, and $c'_j(q) \rightarrow \infty$ as $q \rightarrow \infty$.

Definition 9.3. Partial competitive equilibrium. A price p and quantities $(x_1, \dots, x_I; q_1, \dots, q_J)$ form an equilibrium when

$$\phi'_i(x_i) \leq p, \quad x_i > 0 \Rightarrow \phi'_i(x_i) = p,$$

$$p \leq c'_j(q_j), \quad q_j > 0 \Rightarrow p = c'_j(q_j),$$

and the market for x clears:

$$\sum_i x_i = \sum_j q_j.$$

These are the KKT conditions for consumer and firm optimization plus market clearing.

Equivalently, the unique choices are characterized by

$$x_i(p) = \arg \max_{x \geq 0} \{\phi_i(x) - px\}, \quad q_j(p) = \arg \max_{q \geq 0} \{pq - c_j(q)\}.$$

The associated aggregate excess demand is $z(p) = \sum_i x_i(p) - \sum_j q_j(p)$.

Proposition 9.4. *Existence and uniqueness in the partial model. Suppose $0 < \min_j c'_j(0) < \max_i \phi'_i(0)$ and the preceding curvature and tail assumptions hold. Then z is continuous on the displayed positive-price interval, demand is nonincreasing, and supply is nondecreasing. A clearing price $p^* \in [\min_j c'_j(0), \max_i \phi'_i(0)]$ exists. The maintained strict curvature makes the aggregate crossing strict whenever a side is active, so the equilibrium price is unique.*

To see the sign change, at a price no larger than $\min_j c'_j(0)$ all firms choose zero output while at least one consumer demands a positive amount. At a price at least $\max_i \phi'_i(0)$ all consumers choose zero while at least one firm supplies a positive amount. The intermediate value theorem then gives a clearing price.

The numeraire allocation is recovered from the budget identity

$$m_i = \omega_{mi} + \sum_j \theta_{ij} (pq_j - c_j(q_j)) - px_i.$$

Partial equilibrium deliberately suppresses the wealth effects that would reappear with general preferences or additional markets.

9.3 Welfare in the partial model

Definition 9.5. Pareto efficiency. A feasible allocation is Pareto efficient if no other feasible allocation makes every consumer weakly better off and at least one consumer strictly better off.

Proposition 9.6. *First welfare theorem for the quasilinear model. See [Mas-Colell et al., 1995]. A competitive allocation is Pareto efficient under the stated convexity and local nonsatiation assumptions.*

The planner's problem is

$$\max_{x_i, q_j \geq 0} \left\{ \sum_i \phi_i(x_i) - \sum_j c_j(q_j) \right\} \quad \text{subject to} \quad \sum_i x_i \leq \sum_j q_j.$$

Its KKT multiplier μ satisfies $\phi'_i(x_i) \leq \mu$ with equality for $x_i > 0$ and $\mu \leq c'_j(q_j)$ with equality for $q_j > 0$. Thus a competitive equilibrium satisfies the planner conditions with $\mu = p$.

Proposition 9.7. *Second welfare theorem for the quasilinear model. See [Mas-Colell et al., 1995]. Under concavity, continuity, and the cheaper-bundle condition stated in the general theorem below, any solution of the planner's problem can be decentralized at a supporting price $p = \mu$ by lump-sum transfers of the numeraire whose sum is zero. The theorem concerns allocations or utility vectors attainable by the planner; an arbitrary utility vector need not be feasible.*

9.4 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 9.1 [Core]. Linear partial equilibrium.

Demand and supply are $Q^d(p) = a - bp$ and $Q^s(p) = d(p - c)$ on their economically relevant ranges, with $a, b, d > 0$, $c \geq 0$, and $a > bc$.

- (a) Solve for equilibrium price and quantity.
- (b) Compute consumer and producer surplus using the choke price a/b of inverse demand and the price intercept c of inverse supply, so that the usual triangular-surplus geometry applies.
- (c) Verify that the competitive quantity maximizes total surplus.

Problem 9.2 [Applied]. A per-unit tax.

Introduce a per-unit tax $t > 0$ paid by sellers into the market of Problem 9.1, so buyers pay p_b and sellers receive $p_s = p_b - t$. Assume t is small enough that equilibrium quantity remains nonnegative.

- (a) Solve for p_b , p_s , and quantity.
- (b) Derive tax incidence. Express the local burden shares using the demand and supply slopes, and interpret them using point elasticities evaluated at the common no-tax equilibrium.
- (c) Compute deadweight loss and its derivative with respect to t .

Problem 9.3 [Proof]. Planner and competitive allocation.

A representative consumer has quasilinear utility $u(q) + m$, with u increasing and strictly concave, and production cost $C(q)$ is increasing and convex.

- (a) State and solve the planner's problem using Kuhn-Tucker conditions.
- (b) Show that any interior competitive equilibrium satisfies the same condition.
- (c) State the assumptions required for the converse decentralization claim.

Chapter 10

General Equilibrium Theory

General equilibrium studies all markets simultaneously. Prices determine individual budget sets, firms' feasible transformations, and the wealth generated by ownership claims; an equilibrium is a fixed point at which every agent optimizes and all markets clear.

10.1 Pure exchange economies: the Edgeworth box

Consider two consumers $i = 1, 2$ and two goods $\ell = 1, 2$. Consumer i has consumption set $X_i = \mathbb{R}_+^2$, preference relation $\succeq_i$, and endowment $\omega_i = (\omega_{1i}, \omega_{2i}) \in \mathbb{R}_+^2$. Aggregate endowment is $\omega = \omega_1 + \omega_2 \gg 0$. An allocation $x = (x_1, x_2)$ is feasible when $x_1 + x_2 \leq \omega$ componentwise.

At prices $p \gg 0$, consumer i has wealth $p \cdot \omega_i$ and budget set

$$B_i(p) = \{x_i \in X_i : p \cdot x_i \leq p \cdot \omega_i\}.$$

Definition 10.1. Walrasian equilibrium in the exchange economy. A pair (p^*, x^*) with $p^* \gg 0$ is an equilibrium if markets clear, $x_1^* + x_2^* = \omega$, and

$$x_i^* \succeq_i x_i \quad \text{for every } x_i \in B_i(p^*), \quad i = 1, 2.$$

Prices are defined only up to multiplication by a positive scalar.

Example 10.2. Cobb--Douglas exchange. Suppose $u_i(x_{1i}, x_{2i}) = x_{1i}^\alpha x_{2i}^{1-\alpha}$ with $0 < \alpha < 1$, and endowments $\omega_1 = (1, 2)$ and $\omega_2 = (2, 1)$. The demands are

$$x_1(p) = \left(\frac{\alpha(p_1 + 2p_2)}{p_1}, \frac{(1 - \alpha)(p_1 + 2p_2)}{p_2} \right), \quad x_2(p) = \left(\frac{\alpha(2p_1 + p_2)}{p_1}, \frac{(1 - \alpha)(2p_1 + p_2)}{p_2} \right).$$

Clearing good 1 gives $p_1/p_2 = \alpha/(1 - \alpha)$; one may normalize $p_2 = 1$ and recover the allocation from the two demands.

Definition 10.3. Pareto efficiency. A feasible allocation x is Pareto efficient if there is no feasible x' with $x'_i \succeq_i x_i$ for both consumers and $x'_j \succ_j x_j$ for at least one consumer. The Pareto set is the set of all such allocations for the fixed aggregate endowment.

Proposition 10.4. *First welfare theorem for the exchange box. See [Mas-Colell et al., 1995]. If preferences are locally nonsatiated and (p^*, x^*) is a Walrasian equilibrium, then x^* is Pareto efficient.*

Second welfare theorem for the exchange box. See [Mas-Colell et al., 1995]. If preferences are continuous, convex, and strongly monotone, every Pareto-efficient allocation can be supported by a positive price vector after suitable lump-sum transfers $T_1 + T_2 = 0$. The transfers change wealth, not relative prices.

10.2 One consumer and one producer

There is one consumer with labor-time endowment $\bar{N}$ of time and no initial endowment of the consumption good. Let x_1 be leisure, x_2 consumption, and $f(z)$ output from labor $z \geq 0$. At output price $p > 0$ and wage $w > 0$, the firm solves

$$\max_{z \geq 0} \{pf(z) - wz\}.$$

The consumer owns the firm, receives profit $\pi(p, w)$, and solves

$$\max_{(x_1, x_2) \in X} u(x_1, x_2) \quad \text{subject to} \quad px_2 \leq w(\bar{N} - x_1) + \pi(p, w).$$

Definition 10.5. A competitive equilibrium is a price pair (p^*, w^*) and choices (x^*, z^*) satisfying both optimization problems and the clearing conditions

$$x_2^* = f(z^*), \quad z^* = \bar{N} - x_1^*.$$

Proposition 10.6. *Under continuity, convexity, local nonsatiation, and the usual concavity and*

attainment assumptions, a competitive allocation solves the central-planner problem

$$\max_{x_1, x_2} u(x_1, x_2) \quad \text{subject to} \quad x_2 \leq f(\bar{N} - x_1), \quad 0 \leq x_1 \leq \bar{N}.$$

Conversely, an interior planner solution can be supported by prices equal to the relevant marginal rates.

10.3 Production, ownership, and feasibility

Consider I consumers, J firms, and L commodities. Consumer i has consumption set X_i , preference $\succeq_i$, and endowment $\omega_i \in \mathbb{R}^L$; firm j has production set $Y_j \subseteq \mathbb{R}^L$. Let $\omega = \sum_i \omega_i$. An allocation (x, y) is feasible when

$$\sum_{i=1}^I x_i \leq \omega + \sum_{j=1}^J y_j.$$

In a private-ownership economy, consumer i owns share $\theta_{ij} \in [0, 1]$ of firm j , with $\sum_i \theta_{ij} = 1$ for each j .

Definition 10.7. Walrasian equilibrium. A price vector $p \gg 0$ and allocation (x^*, y^*) form an equilibrium when, for every firm,

$$y_j^* \in \arg \max_{y_j \in Y_j} p \cdot y_j,$$

and, for every consumer,

$$x_i^* \in \arg \max_{x_i \in X_i} \{u_i(x_i) : p \cdot x_i \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*\},$$

with market clearing $\sum_i x_i^* = \omega + \sum_j y_j^*$.

Definition 10.8. Price equilibrium with transfers. Replace each consumer's ownership wealth by an assigned w_i satisfying $\sum_i w_i = p \cdot \omega + \sum_j p \cdot y_j^*$, and require $x_i^* \in \arg \max \{u_i(x_i) : p \cdot x_i \leq w_i\}$ while firms maximize profit and markets clear.

10.4 Welfare theorems

Proposition 10.9. *First Fundamental Welfare Theorem.* See [Debreu, 1959, Mas-Colell et al., 1995]. If preferences are locally nonsatiated and (p, x^*, y^*) is a price equilibrium with transfers, then (x^*, y^*) is Pareto efficient. A Walrasian equilibrium is the special case in which wealth is generated by endowments and ownership shares.

The proof is the supporting-price argument. Consumer optimality and local nonsatiation imply that every bundle weakly preferred to x_i^* costs at least w_i ; a strictly preferred bundle costs strictly more. A Pareto improvement therefore has weakly greater expenditure for every consumer and strictly greater expenditure for at least one consumer. Summing and combining this strict inequality with firm profit maximization contradicts feasibility and market clearing.

Definition 10.10. Price quasi-equilibrium with transfers. A nonzero price vector, allocation, and assigned wealths form a price quasi-equilibrium with transfers when firms maximize profit, markets clear, $p \cdot x_i^* = w_i$ for every consumer, the assigned wealths add to aggregate endowment and profit value, and every bundle strictly preferred to x_i^* has cost at least w_i .

Proposition 10.11. *Second Fundamental Welfare Theorem. See [Debreu, 1959, Mas-Colell et al., 1995]. Suppose every production set is closed and convex, every preference is convex, continuous, and locally nonsatiated, and the aggregate attainable set is closed. Every Pareto-efficient allocation can be supported by a nonzero price vector as a price quasi-equilibrium with transfers. If, for every consumer, there is a bundle $\hat{x}_i \in X_i$ with $p \cdot \hat{x}_i < w_i$, continuity and convexity upgrade the quasi-equilibrium to a price equilibrium with transfers.*

The result follows by separating the convex aggregate set of bundles that make every consumer strictly better off from the convex aggregate production-plus-endowment set. The existence of a strictly cheaper feasible bundle for each consumer supplies the final upgrade from weak support to full budget maximization.

10.5 General equilibrium with inattention

An inattention model must distinguish true prices from the prices or signals used by a consumer's decision rule. Let $M_i \in \mathbb{R}^{L \times L}$ be an attention parameter and let $D_i(p, M_i)$ be the demand correspondence generated by the fully specified behavioral model. The model must state whether M_i changes perceived prices, the budget, utility, or information.

For a price-perception benchmark, $\tilde{p}_i = M_i p$ and a possible behavioral budget are written explicitly, for example

$$D_i(p, M_i) = \arg \max_{x \in X_i} \{u_i(x) : \tilde{p}_i \cdot x \leq \tilde{p}_i \cdot \omega_i\}.$$

This is not a Walrasian budget unless $M_i = I_L$ or the model separately establishes equivalence with the true budget.

Definition 10.12. Inattention equilibrium. A pair (p, x) is an inattention equilibrium when each $x_i \in D_i(p, M_i)$ satisfies the model's actual feasibility restrictions and

$$\sum_{i=1}^I x_i = \sum_{i=1}^I \omega_i.$$

Proposition 10.13. *Compatibility with Pareto efficiency. Suppose preferences are differentiable, strictly increasing, and concave and x^* is an interior Pareto-efficient allocation. Then there is a common supporting price vector $q \gg 0$ and multipliers $\mu_i > 0$ such that $\nabla u_i(x_i^*) = \mu_i q$ for every i . If the behavioral first-order condition is $\nabla u_i(x_i^*) = \lambda_i M_i p$ with $\lambda_i > 0$, then each $M_i p$ must be collinear with the same q .*

This is a necessary compatibility condition, not an if-and-only-if statement about attention matrices. Identical matrices can support efficiency in a regular model, while different matrices can also be compatible at special allocations. Perceived-price optimality alone does not justify either welfare theorem.

10.6 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 10.1 [Core]. Cobb-Douglas exchange equilibrium.

There are two consumers with common utility $u_i(x_{1i}, x_{2i}) = x_{1i}^\alpha x_{2i}^{1-\alpha}$, $0 < \alpha < 1$, and endowments $\omega_1 = (1, 2)$, $\omega_2 = (2, 1)$.

- Derive individual Walrasian demands.
- Normalize $p_2 = 1$ and solve for the equilibrium relative price.
- Compute equilibrium allocations and verify both market-clearing equations.

Problem 10.2 [Proof]. First Fundamental Welfare Theorem.

Consider a private-ownership economy with locally nonsatiated preferences and profit-maximizing firms.

- State the definition of Pareto efficiency used in the theorem.
- Prove that a competitive equilibrium allocation is Pareto efficient.
- Identify exactly where local nonsatiation, budget feasibility, profit maximization, and market clearing enter the proof.

Problem 10.3 [Proof]. Supporting a Pareto optimum.

Suppose consumption sets and production sets are closed and convex, preferences are continuous, convex, and locally nonsatiated, and a feasible allocation is Pareto efficient. Assume the separation hypotheses stated in the chapter.

(a) Formulate the aggregate set-separation argument and show how it supports the allocation as a price quasi-equilibrium with transfers.

(b) Explain why the initial endowment distribution generally must be changed by lump-sum transfers.

(c) State the positive-wealth or boundary condition needed to upgrade the quasi-equilibrium to a price equilibrium with transfers, and distinguish that object from a competitive equilibrium for the original ownership economy.

Problem 10.4 [Applied]. One consumer and one producer.

There are two goods. Good 1 can be consumed or used as input z to produce good 2 according to $y_2 = \sqrt{z}$. The consumer owns the firm and has an endowment $\bar{\omega}_1 > 0$ of good 1 and no good

2. Utility is $\log x_1 + \log x_2$.

(a) Solve the planner's problem.

(b) Find supporting relative prices from marginal conditions.

(c) Verify that the consumer and firm choices clear both markets.

Chapter 11

Positive Theory of Equilibrium

The preceding chapter established welfare properties of competitive equilibrium. This chapter studies what those conditions imply—and what they leave undetermined—for aggregate excess demand, existence, local uniqueness, comparative statics, and the core.

11.1 Aggregate excess demand in pure exchange

Consider a pure-exchange economy with consumers $i = 1, \dots, I$, commodity space $\mathbb{R}_+^L$, endowments $\omega_i \in \mathbb{R}_+^L$, and aggregate endowment $\omega = \sum_i \omega_i \gg 0$. At a strictly positive price vector $p \gg 0$, consumer i has wealth $w_i = p \cdot \omega_i$ and demand correspondence

$$D_i(p, w_i) = \arg \max \{u_i(x) : x \geq 0, p \cdot x \leq w_i\}.$$

Definition 11.1. Aggregate excess demand. With possibly multivalued demand, define the Minkowski-sum correspondence

$$Z(p) = \sum_{i=1}^I D_i(p, p \cdot \omega_i) - \{\omega\}.$$

When every demand is single-valued, write its unique selection as $x_i(p)$ and the resulting excess-demand function as $z(p) = \sum_i x_i(p) - \omega$.

Pure-exchange equilibrium. A pair (p^*, x^*) with $p^* \gg 0$ is a Walrasian equilibrium when each $x_i^* \in D_i(p^*, p^* \cdot \omega_i)$ and $\sum_i x_i^* = \omega$. Prices are equivalent up to multiplication by a positive scalar.

Proposition 11.2. *Walras' law and homogeneity. Suppose each consumer is locally nonsatiated*

and demands are single-valued. Then

$$p \cdot z(p) = 0, \quad z(\lambda p) = z(p) \quad (\lambda > 0).$$

If $p \gg 0$ and markets clear in all but one commodity, Walras' law implies that the final market clears as well.

If preferences are continuous, strictly convex, and strongly monotone, the demand function is continuous on $\mathbb{R}_{++}^L \times \mathbb{R}_{++}$. Consequently z is continuous on strictly positive prices. Its domain should not be silently extended to a zero price: demand may become unbounded there. In an equilibrium with nonnegative prices, a good can have excess supply only at a zero price; under strong monotonicity and $p \gg 0$, every market clears.

For calculations it is convenient to normalize prices. We use either the simplex

$$\Delta = \{p \in \mathbb{R}_+^L : \mathbf{1} \cdot p = 1\}$$

or the chart $p_L = 1$. These are equivalent for strictly positive prices.

Example 11.3. Cobb--Douglas exchange. With $u_i(x) = x_1^\alpha x_2^{1-\alpha}$ and $0 < \alpha < 1$,

$$x_i(p) = \left(\frac{\alpha p \cdot \omega_i}{p_1}, \frac{(1-\alpha) p \cdot \omega_i}{p_2} \right).$$

Substitution into $z(p) = 0$ reduces equilibrium to one relative-price equation, because the second equation follows from Walras' law.

11.2 Existence

Proposition 11.4. *Walrasian existence theorem. See [Arrow and Debreu, 1954, Debreu, 1959]. Consider a finite pure-exchange economy with consumption sets $\mathbb{R}_+^L$, individual endowments $\omega_i \gg 0$, and complete, transitive, continuous, convex, and strongly monotone preferences. Then a Walrasian equilibrium with $p^* \gg 0$ exists. Strict convexity is not required for existence; it makes each consumer's demand single-valued.*

The proof normalizes prices to Δ , truncates consumption at the compact aggregate feasible set, constructs a correspondence with nonempty compact convex values, and applies a fixed-point theorem. Strictly positive individual endowments provide positive wealth at every nonzero price, and strong monotonicity rules out a zero price for any commodity in equilibrium. These

hypotheses supply the boundary step that a fixed-point map defined only on $\mathbb{R}_{++}^L$ would otherwise omit.

11.3 Local uniqueness and regularity

Definition 11.5. Normalized local uniqueness. A normalized equilibrium $p^* \in \Delta$ is locally unique if some neighborhood of p^* contains no other normalized equilibrium.

Work in the chart $p_L = 1$ and write $\widehat{z}(p_1, \dots, p_{L-1}) = (z_1, \dots, z_{L-1})$. The last component is redundant by Walras' law.

Definition 11.6. Regular equilibrium. A differentiable equilibrium p^* is regular when the Jacobian $D\widehat{z}(p^*)$ is nonsingular.

Proposition 11.7. *Regularity implies local uniqueness. See [Mas-Colell et al., 1995]. If $D\widehat{z}(p^*)$ is nonsingular, the inverse function theorem implies that p^* is an isolated zero of $\widehat{z}$. This is a local statement; it does not rule out another equilibrium far away.*

Fix the chart $p_L = 1$ with its standard orientation. For a regular economy, define the signed index of an equilibrium as $\text{sign det}[-D\widehat{z}(p^*)]$. The index theorem [Mas-Colell et al., 1995] states that when the normalized equilibrium set is compact and excess demand points inward on the boundary of the chosen compact price region, the sum of these indices is one. Hence a regular economy has an odd number of equilibria. The conclusion is global and does not follow from the inverse function theorem alone.

11.4 What equilibrium theory does not identify

At a differentiability point, Walras' law and homogeneity imply the exact derivative restrictions

$$D_p z(p)^\top p = -z(p), \quad D_p z(p)p = 0.$$

At an equilibrium, the first identity becomes $D_p z(p)^\top p = 0$; these are restrictions, not a complete theory of comparative statics.

Proposition 11.8. *Sonnenschein--Mantel--Debreu theorem. See [Sonnenschein, 1972, Mantel, 1974, Debreu, 1974]. Subject to continuity, homogeneity of degree zero, Walras' law, and boundary divergence toward any zero-priced desirable good, aggregate excess demand can have essentially*

arbitrary shape on compact subsets of the strictly positive normalized price simplex. In particular, individual rationality does not generally imply uniqueness, stability, or monotone comparative statics for the aggregate equilibrium map.

The theorem is an impossibility result about restrictions on aggregate demand. It does not say that every arbitrary function is a demand system: the stated regularity, Walras, homogeneity, and boundary conditions remain necessary, and the result concerns a constructed finite exchange economy.

11.5 Uniqueness and gross substitutes

With a constant-returns production cone $Y \subseteq \mathbb{R}^L$ containing zero, a strictly positive price p supports the technology when $p \cdot y \leq 0$ for every $y \in Y$. If aggregate excess demand satisfies $z(p) \in Y$ and Walras' law, then choosing production $y = z(p)$ clears markets and maximizes profit. This observation separates the consumer condition from the technology-support condition.

Definition 11.9. Gross substitutes. An excess-demand function has the strict gross-substitutes property if, whenever only the price of good ℓ rises, excess demand for every other good $k \neq \ell$ strictly rises. Weak gross substitutes replaces strict inequalities with weak inequalities. The property is imposed on the relevant price domain and is stronger than Walras' law or homogeneity.

Proposition 11.10. *Gross-substitutes uniqueness theorem. See [Mas-Colell et al., 1995]. Under continuity, homogeneity, Walras' law, and a strict gross-substitutes condition, a pure-exchange equilibrium has at most one normalized strictly positive price vector. Existence still requires a separate existence theorem. Without strictness, gross substitutes can deliver a convex or lattice-structured equilibrium set rather than a unique price.*

11.6 The core

For clarity, define the core first in a pure-exchange economy. There are I consumers, endowments $\omega_i \in \mathbb{R}_+^L$, and a feasible allocation satisfies $\sum_i x_i = \sum_i \omega_i$ (free disposal can replace equality by $\leq$). A nonempty coalition $S \subseteq \{1, \dots, I\}$ can use only $\sum_{i \in S} \omega_i$.

Definition 11.11. Blocking coalition. Coalition S blocks a feasible allocation x if there are $\{x'_i\}_{i \in S}$ with $x'_i \succ_i x_i$ for every $i \in S$ and $\sum_{i \in S} x'_i \leq \sum_{i \in S} \omega_i$.

The core is the set of feasible allocations blocked by no coalition.

Proposition 11.12. *Walrasian allocations are in the core. See [Debreu and Scarf, 1963, Mas-Colell et al., 1995]. Suppose preferences are locally nonsatiated and (p^*, x^*) is a Walrasian equilibrium with $p^* \gg 0$. If a coalition made every member strictly better off, each member would need a bundle with strictly greater value than $p^* \cdot \omega_i$. Summing over the coalition contradicts its resource constraint. Thus no coalition blocks x^* .*

The converse need not hold in a finite economy: the core can contain noncompetitive allocations. In the r -fold replica of a fixed finite economy, identify equal-treatment allocations with their per-type consumption vectors. Under continuous, convex, locally nonsatiated preferences and strictly positive aggregate resources, the Debreu--Scarf core-convergence theorem [Debreu and Scarf, 1963] says that every limit point of such normalized core allocations as $r \rightarrow \infty$ is a Walrasian allocation; conversely, every Walrasian allocation belongs to the core of every replica.

11.7 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 11.1 [Core]. Differential identities for excess demand.

Let aggregate excess demand $z : \mathbb{R}_{++}^L \rightarrow \mathbb{R}^L$ be continuously differentiable, homogeneous of degree zero, and satisfy Walras' law.

- (a) Differentiate homogeneity to show $D_p z(p)p = 0$.
- (b) Differentiate Walras' law to show $D_p z(p)^\top p = -z(p)$.
- (c) Specialize the second identity to an equilibrium and interpret the left null vector.

Problem 11.2 [Proof]. Regularity and local uniqueness.

Normalize prices by $p_L = 1$ and let $\hat{z}$ be the first $L - 1$ excess-demand equations.

- (a) Define a regular equilibrium.
- (b) Use the inverse function theorem to prove that a regular equilibrium is locally isolated.
- (c) Explain why local isolation implies finitely many equilibria only after compactness and boundary conditions are added.

Problem 11.3 [Proof]. Gross substitutes and uniqueness.

Assume aggregate excess demand satisfies strict gross substitutes on the relevant price domain: increasing one price strictly raises excess demand for every other good whenever the comparison is nondegenerate. Suppose two strictly positive normalized equilibrium price vectors exist.

- (a) Compare the largest relative price change across goods.
- (b) Use gross substitutes and Walras' law to derive a contradiction unless the two price vectors are proportional.
- (c) Explain why weak gross substitutes alone need not imply uniqueness and where strictness (or an irreducibility substitute) enters the proof.

Problem 11.4 [Core]. Competitive allocations and the strong core.

Use the strong-blocking convention in a finite pure-exchange economy with locally nonsatiated preferences: a coalition blocks when all its members can be made weakly better off using their own aggregate endowment and at least one member is made strictly better off. Define the associated strong core.

- (a) Prove that every Walrasian allocation belongs to the strong core.
- (b) Give an economic reason why the converse can fail with finitely many consumers.
- (c) State the content, but not a proof, of the core-convergence theorem for replica economies.

Part V

Advanced Equilibrium Applications

Chapter 12

Testable Restrictions on the Equilibrium Manifold

This chapter studies what a finite set of market observations can reveal about a pure-trade economy. The central tension is between individual utility maximization and market clearing: each consumer must rationalize their own choices, while the choices must add up to the observed aggregate endowment.

12.1 Observables and the equilibrium manifold

Fix a finite set of consumers $i = 1, \dots, I$, goods $\ell = 1, \dots, L$, and observations $t = 1, \dots, T$. At observation t , let $p^t \in \mathbb{R}_{++}^L$ denote the price vector, let $w_i^t > 0$ denote consumer i 's income, and let $\omega^t \in \mathbb{R}_+^L$ denote the aggregate endowment. The accounting restriction is $\sum_i w_i^t = p^t \cdot \omega^t$. Prices are normalized because only their positive scale is economically relevant.

Definition 12.1. (Pure-trade rationalization.) A finite dataset is rationalized by a pure-trade economy if there are continuous, locally nonsatiated, concave utility functions u_i and bundles $x_i^t \in \mathbb{R}_+^L$ such that, for every i, t , $x_i^t \in \arg \max\{u_i(x) : p^t \cdot x \leq w_i^t\}$, and, for every t , $\sum_i x_i^t = \omega^t$.

(Equilibrium manifold.) For a market excess-demand function $z(\hat{\omega}, p)$, where $\hat{\omega} = (\omega_1, \dots, \omega_I)$, the equilibrium manifold is $\mathcal{M} = \{(\hat{\omega}, p) : z(\hat{\omega}, p) = 0\}$. It records pairs of latent individual-endowment profiles and prices that can clear markets for fixed underlying preferences. The Brown--Matzkin observable set below is a projection of this theoretical manifold onto prices, individual incomes, and aggregate endowments.

12.2 Afriat inequalities and elimination of unobservables

For a finite collection of observed or latent bundles with strictly positive prices and expenditure exhaustion, Afriat's inequalities are

$$u_i^t \leq u_i^s + \lambda_i^s p^s \cdot (x_i^t - x_i^s) \quad \text{for all } s, t,$$

with $\lambda_i^s > 0$. They are equivalent to the existence of a monotone, concave utility representation for the finite choice data.

The equilibrium restrictions combine these inequalities with the budget equalities and market-clearing equalities

$$p^t \cdot x_i^t = w_i^t, \quad \sum_i x_i^t = \omega^t.$$

The unobserved variables are

$$(x_i^t, u_i^t, \lambda_i^t)_{i,t}.$$

Prices, individual incomes, and aggregate endowments are the data. Because λ_i^s multiplies latent x_i^t , the joint system is polynomial but not linear in all unknowns.

(Tarski--Seidenberg theorem.) The projection of a semialgebraic set is semialgebraic. Consequently, eliminating the unobserved variables from any finite polynomial system yields a quantifier-free semialgebraic condition in the observables. This theorem supplies an elimination principle; it does not say that the resulting restrictions are easy to compute or that every model has a nontrivial restriction.

12.3 Brown--Matzkin theorem

Brown and Matzkin (1996) apply this principle to pure-trade equilibrium. Under their regularity and invariance assumptions, finite observations on prices, individual incomes, and aggregate endowments are rationalizable if and only if they satisfy a finite family of semialgebraic restrictions obtained after eliminating the latent utility levels, marginal utilities of income, and consumption bundles. The restrictions are nontrivial: some finite datasets cannot lie on the equilibrium manifold of any economy in the maintained class. See [Brown and Matzkin, 1996]; its quantifier-elimination construction is not reproved in this text.

The theorem should not be confused with the Sonnenschein--Mantel--Debreu result. The latter concerns the flexibility of aggregate excess demand when individual preferences are allowed to

vary; Brown--Matzkin hold individual preferences fixed across observations and exploit variation in observed incomes and aggregate endowments.

12.4 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 12.1 [Core]. Equilibrium Afriat feasibility.

At each observation t , prices $p^t \gg 0$, individual incomes $(w_i^t)_i$, and aggregate endowment ω^t are observed, while individual allocations are latent.

(a) Introduce latent consumptions x_i^t and write individual budget equalities $p^t \cdot x_i^t = w_i^t$ and aggregate market clearing $\sum_i x_i^t = \omega^t$.

(b) Write Afriat inequalities for every consumer across observations.

(c) Explain why joint feasibility is a finite semialgebraic revealed-equilibrium test, why the latent multipliers and bundles make the full system bilinear, and how quantifier elimination projects the restrictions onto the observables.

Problem 12.2 [Proof]. What equilibrium data identify.

Suppose only equilibrium prices and aggregate endowments are observed.

(a) List the preference, allocation, and ownership objects that remain unobserved.

(b) Explain why market clearing alone cannot identify individual demand.

(c) Describe how additional observations, known individual endowments, or parametric restrictions can turn an underidentified model into a testable one.

Problem 12.3 [Applied]. Counterfactual discipline.

A dataset passes a revealed-equilibrium feasibility test under a maintained class of preferences.

(a) Explain why passing the test does not point-identify a counterfactual equilibrium.

(b) Define an identified set for a counterfactual price or allocation.

(c) State two economically meaningful restrictions that could tighten this set, and explain the cost of each restriction.

Chapter 13

Sequential Trade

13.1 Arrow--Debreu benchmark

There are S states and L goods in each state. Consumer i has state-contingent endowment $\omega_i = (\omega_{i1}, \dots, \omega_{iS})$ and preference $\succeq_i$ over contingent consumption plans $x_i = (x_{i1}, \dots, x_{iS})$.

Definition 13.1. (Arrow--Debreu equilibrium.) Prices $p = (p_1, \dots, p_S) \in (\mathbb{R}_{++}^L)^S$ and an allocation $x^* = (x_1^*, \dots, x_I^*)$ form an Arrow--Debreu equilibrium if, for each consumer, x_i^* maximizes $\succeq_i$ over $B_i^{AD}(p) = \{x_i : \sum_{s=1}^S p_s \cdot x_{is} \leq \sum_{s=1}^S p_s \cdot \omega_{is}\}$, and markets clear state by state: $\sum_i x_{is}^* = \sum_i \omega_{is}$ for every s .

13.2 Spot markets and sequential trade

Suppose contingent commodity markets are unavailable at date $t = 0$. There are K tradable assets with payoff matrix $R \in \mathbb{R}^{S \times K}$: one unit of asset k pays R_{sk} units of the first good in state s . Asset prices are $q \in \mathbb{R}^K$, and spot-good prices in state s are $p_s \in \mathbb{R}_{++}^L$.

Definition 13.2. (Radner budget set.) Given asset holdings $z_i \in \mathbb{R}^K$ and state-contingent consumption x_i , consumer i satisfies $q \cdot z_i \leq 0$ and $p_s \cdot x_{is} \leq p_s \cdot \omega_{is} + p_{s1}(Rz_i)_s$ for every s . The zero on the date-0 constraint is the normalization of zero initial asset endowment.

(Radner equilibrium.) A price system $(q, p_1, \dots, p_S)$ and plans $(z_i^*, x_i^*)_{i=1}^I$ form a Radner equilibrium when every consumer maximizes their preference over the Radner budget set, asset markets clear $\sum_i z_i^* = 0$, and goods markets clear in every state: $\sum_i x_{is}^* = \sum_i \omega_{is}$ for all s .

Proposition 13.3. (Arrow--Radner equivalence.) See [Radner, 1972, Mas-Colell et al., 1995]. Suppose consumption lies in $\mathbb{R}_+^{LS}$, preferences are continuous and strictly monotone, the payoff

matrix has full row rank $\text{rank}(R) = S$, and asset prices satisfy no arbitrage. Full spanning and no arbitrage yield a strictly positive state-price vector ψ with $q = R^\top \psi$. For positive spot prices, define Arrow--Debreu commodity prices by $\tilde{p}_s = (\psi_s/p_{s1})p_s$. Then the consumer budget sets project to the same contingent-consumption budget, so the sets of consumption allocations supported by Arrow--Debreu and Radner equilibria coincide. If $\text{rank}(R) < S$, some contingent trades are unattainable and the equivalence need not hold.

The full-rank condition is the precise meaning of complete asset markets in this finite-state model. The proposition is an allocation-equivalence statement conditional on equilibrium prices; it does not by itself assert equilibrium existence.

13.3 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 13.1 [Core]. Spanning contingent consumption.

There are S states tomorrow and K traded assets with payoff matrix $R \in \mathbb{R}^{S \times K}$ and date-zero asset-price vector $q \in \mathbb{R}^K$. Assume no arbitrage.

- (a) State the rank condition under which every contingent net trade can be implemented by an asset portfolio.
- (b) If $S = 2$ and the assets have payoff vectors $(1, 1)$ and $(1, 0)$, construct the portfolio that delivers any target payoff (c_1, c_2) .
- (c) Let $\psi \in \mathbb{R}_{++}^S$ be state prices satisfying $q = R^\top \psi$. Explain how the date-zero portfolio budget maps into an Arrow-Debreu present-value budget.

Problem 13.2 [Core]. Incomplete markets.

Let $S = 3$ and suppose the only assets are a bond with payoff $(1, 1, 1)$ and an asset with payoff $(1, 0, 0)$.

- (a) Characterize the attainable payoff subspace.
- (b) Determine whether the claim $(0, 1, -1)$ is attainable.
- (c) Explain why competitive equilibrium with these assets need not be Pareto efficient relative to the full contingent-commodity economy.

Problem 13.3 [Proof]. Arrow-Debreu and sequential equivalence.

There are S states, K assets with payoff matrix $R \in \mathbb{R}^{S \times K}$ of rank S , date-zero asset prices q , and state- s spot-price vectors p_s . Asset payoffs are denominated in the state numeraire. Consumer i chooses a portfolio z_i subject to $q \cdot z_i \leq 0$, then chooses x_{is} subject to $p_s \cdot x_{is} \leq p_s \cdot \omega_{is} + p_s 1(Rz_i)_s$. Assume no arbitrage and strictly monotone continuous preferences.

- (a) Map an Arrow-Debreu budget into date-zero asset purchases and state-by-state spot trades.
- (b) Prove the reverse mapping from a sequential budget to a present-value budget.
- (c) Identify which step fails under incomplete markets or binding portfolio constraints.

Chapter 14

Matching

14.1 One-to-one matching

Let M and W be finite sets of agents. Each agent has a strict preference ordering over the agents on the other side and the option of remaining unmatched. A matching μ assigns each agent either an acceptable partner or themselves, and satisfies $\mu(\mu(a)) = a$.

Definition 14.1. (Stability.) A matching is individually rational if no agent is matched with an unacceptable partner. A pair (m, w) not matched to one another blocks μ if $w \succ_m \mu(m)$ and $m \succ_w \mu(w)$. A matching is stable when it is individually rational and has no blocking pair.

14.2 Deferred acceptance

In the men-proposing deferred-acceptance algorithm, each unmatched man proposes to the most-preferred woman who has not rejected him. Each woman tentatively holds her most-preferred acceptable proposal received so far and rejects the rest. The process stops when no proposal is rejected.

Theorem 14.2. (*Gale--Shapley theorem.*) See [Gale and Shapley, 1962]. For every finite marriage problem, deferred acceptance terminates and returns a stable matching. Among all stable matchings, the outcome is weakly best for every man and weakly worst for every woman.

Termination is finite because a man proposes to each woman at most once. The stability and proposer-optimality statements are the Gale--Shapley theorem.

14.3 Housing markets and top trading cycles

In a housing market, each agent initially owns one house and has a strict preference over houses. The strong core consists of allocations for which no coalition can make every member weakly better off and at least one member strictly better off using only the houses owned by that coalition.

The top-trading-cycles algorithm directs each agent to point to their favorite remaining house and each house to point to its owner. Every directed cycle is executed, the participating houses and agents are removed, and the procedure is repeated.

Theorem 14.3. (*Top-trading-cycles theorem.*) See [Shapley and Scarf, 1974]. With strict preferences and one-house ownership, top trading cycles terminates, is Pareto efficient, and returns the unique allocation in the strong core; the same allocation is supported as the competitive allocation of the housing market.

14.4 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 14.1 [Core]. Deferred acceptance by hand.

Three men have rankings $m_1 : w_1 \succ w_2 \succ w_3$, $m_2 : w_1 \succ w_2 \succ w_3$, and $m_3 : w_2 \succ w_3 \succ w_1$.

The women rank $w_1 : m_2 \succ m_1 \succ m_3$, $w_2 : m_1 \succ m_3 \succ m_2$, and $w_3 : m_3 \succ m_2 \succ m_1$.

- (a) Run men-proposing deferred acceptance round by round.
- (b) Verify stability of the resulting matching by checking potential blocking pairs.
- (c) Explain why termination is guaranteed in every finite market.

Problem 14.2 [Proof]. Proposer optimality.

Consider a finite balanced one-to-one matching market in which every participant has a complete strict ranking of the other side and all partners are acceptable.

- (a) Prove that if a receiver rejects a proposer during deferred acceptance, that rejected proposer-receiver pair cannot be matched to each other in any stable matching.
- (b) Use this rejection lemma to prove proposer optimality among stable matchings.
- (c) Explain why the receiver side obtains its least-preferred stable matching.

Problem 14.3 [Core]. Top trading cycles.

Agents 1, 2, 3 initially own houses h_1, h_2, h_3 . Their rankings are 1 : $h_2 \succ h_1 \succ h_3$, 2 : $h_1 \succ h_2 \succ h_3$, and 3 : $h_1 \succ h_3 \succ h_2$.

- (a) Run top trading cycles.
- (b) Show that the outcome is individually rational and Pareto efficient.
- (c) Verify directly that no coalition blocks the outcome.

Chapter 15

Computable General Equilibrium

15.1 From accounts to counterfactuals

Computable general equilibrium (CGE) analysis uses a calibrated microeconomic model and a social accounting matrix to evaluate policy counterfactuals. Calibration chooses parameters so that the model reproduces a benchmark allocation and the associated accounting identities before the policy is changed.

15.2 A two-sector small open economy

The benchmark economy has two goods, aggregate labor N , and sector-specific capital K_j . A small open economy takes world prices as given. For sector j , a constant-elasticity-of-substitution production function is

$$Q_j = A_j \left[\alpha_j K_j^{-\rho_j} + (1 - \alpha_j) N_j^{-\rho_j} \right]^{-1/\rho_j},$$

with $A_j > 0$, $0 < \alpha_j < 1$, and $\rho_j > -1$, $\rho_j \neq 0$. The associated elasticity parameter is $\sigma_j = 1/(1 + \rho_j) > 0$.

Definition 15.1. (Cost and conditional factor demand.) Given output $\bar{Q}_j$ and factor prices (r_j, W) , the unit-consistent cost function is

$$c_j(r_j, W, \bar{Q}_j) = \min_{K_j, N_j \geq 0} \{r_j K_j + W N_j : Q_j(K_j, N_j) \geq \bar{Q}_j\}.$$

When differentiable, conditional factor demands satisfy

$$K_j^D = \frac{\partial c_j}{\partial r_j}, \quad N_j^D = \frac{\partial c_j}{\partial W}.$$

This derivative formulation avoids confusing the price P_j with the CES parameter ρ_j .

15.3 Social accounting and calibration

The social accounting matrix records production, factor payments, household income, and expenditure flows that must balance at the benchmark. For a Cobb–Douglas expenditure equation $P_j Q_{h,j}^D = \beta_{h,j} Y_h$, the observed benchmark share identifies $\beta_{h,j} = P_j Q_{h,j}^D / Y_h$. The calibrated model is checked by substituting the parameter values back into every accounting identity and market-clearing condition.

15.4 Equilibrium closure

A computable model is not defined by technology and accounting identities alone. Let households be indexed by h and sectors by j . Household income must specify factor ownership, taxes, transfers, and the disposition of profits and tariff revenue; a representative identity is

$$Y_h = W \bar{N}_h + \sum_j r_j \bar{K}_{hj} + T_h + \Pi_h - \text{Tax}_h.$$

Demand equations $x_{hj} = x_{hj}(P, Y_h)$ then close the household side.

For each good, the resource balance distinguishes domestic output, imports, exports, intermediate use, household demand, investment, and government demand:

$$Q_j + M_j = \sum_h x_{hj} + G_j + I_j + X_j + \sum_k a_{jk} Q_k.$$

Factor markets satisfy

$$\sum_j N_j = \sum_h \bar{N}_h, \quad K_j = \sum_h \bar{K}_{hj} \quad \text{for sector-specific capital.}$$

Constant returns and competitive pricing impose zero profit, so the domestic producer price equals unit cost whenever the sector operates.

The government budget identifies the use of tax and tariff receipts, and the external account

identifies the financing of net imports. A balanced-trade closure, for example, imposes

$$\sum_j P_j^W M_j = \sum_j P_j^W X_j,$$

whereas a fixed foreign transfer replaces the right-hand side by exports plus that transfer. The model must choose one nominal normalization, such as $W = 1$ or $ER = 1$, and must not impose both as independent equilibrium equations.

A benchmark replication check solves this entire system at the calibrated parameters and reports the maximum absolute residual in every household, firm, market, government, and external-account equation. Counterfactual welfare must also be declared, for example equivalent variation relative to the benchmark utility and price system.

15.5 Counterfactual policies

For an imported good i , let ER denote the exchange rate and P_i^W the world price. A tariff rate $\tau_i \geq 0$ changes the domestic price according to

$$P_i = ER P_i^W (1 + \tau_i).$$

Free trade is the baseline $\tau_i = 0$; a tariff counterfactual changes τ_i and resolves the full equilibrium system. Other counterfactuals change tax, subsidy, transfer, or technology equations while leaving the accounting identities and calibrated preference and technology parameters in place.

15.6 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem 15.1 [Core]. Calibrating a CES technology.

A constant-returns technology is $Q = A[\alpha K^{-\rho} + (1 - \alpha)\ell^{-\rho}]^{-1/\rho}$, with $\rho > -1$ and $\rho \neq 0$. A strictly positive interior cost-minimizing benchmark observation gives (Q, K, ℓ, r, W) and an elasticity of substitution $\sigma > 0$, $\sigma \neq 1$. (The Cobb-Douglas limit $\sigma = 1$ is outside the displayed parameterization.)

- (a) Recover ρ from σ .

- (b) Use the cost-minimizing first-order condition to recover α .
- (c) Recover A and verify that the benchmark is exactly replicated.

Problem 15.2 [Core]. Social-accounting consistency.

A two-sector social accounting matrix records sectoral sales, intermediate purchases, labor and capital payments, household income, taxes, savings, and imports.

- (a) State the row-balance and column-balance conditions.
- (b) Explain why every market-clearing equation has a corresponding accounting identity but not every accounting identity is a behavioral equation.
- (c) Design a numerical residual check that would detect an incorrectly calibrated benchmark.

Problem 15.3 [Applied]. A tariff counterfactual.

A small open economy takes the world price P^W and exchange rate ER as given. A tariff τ implies domestic price $P = \text{ER } P^W (1 + \tau)$. Let differentiable domestic demand and supply be $D(P)$ and $S(P)$, imports be $M(P) = D(P) - S(P) > 0$, and tariff revenue be returned lump-sum, $\text{TR}(\tau) = \tau \text{ER } P^W M(P(\tau))$. Interpret D as quasi-linear (or otherwise free of income effects) and S as competitive supply generated by marginal cost, so consumer and producer surplus are valid partial-equilibrium welfare measures.

- (a) Derive the first-order change in the domestic price from a small tariff change.
- (b) Derive the symbolic first-order decomposition $d\mathcal{W} = d\text{CS} + d\text{PS} + d\text{TR}$ and simplify it at free trade. Do not assume numerical functional forms.
- (c) Explain which general-equilibrium channels are omitted by that approximation and must be restored in a CGE counterfactual.

Appendix A

Order Theory

This mathematical appendix collects the order-theoretic concepts used throughout the book. The orientation is economic: xRy means that x is ranked at least as high as y . Unless stated otherwise, X is a nonempty set.

A.1 Binary Relations

Definition A.1. A binary relation on X is a subset $R \subseteq X \times X$. We write xRy in place of $(x, y) \in R$. The inverse relation R^{-1} is defined by $xR^{-1}y \iff yRx$.

A.1.1 Properties

A relation R on X is:

Reflexive if xRx for every $x \in X$.

Complete (or total) if, for every $x, y \in X$, either xRy or yRx (or both). Completeness implies reflexivity by taking $x = y$.

Symmetric if xRy implies yRx .

Asymmetric if xRy implies $\neg(yRx)$. Every asymmetric relation is irreflexive.

Antisymmetric if xRy and yRx imply $x = y$. Antisymmetry does not mean the negation of symmetry.

Transitive if xRy and yRz imply xRz .

Acyclic if there do not exist $n \geq 1$ and $x_0, \dots, x_n \in X$ such that $x_0 = x_n$ and $x_{k-1}Rx_k$ for every $k = 1, \dots, n$. Acyclicity is normally imposed on a strict relation.

Example A.2. The equality relation on X is reflexive, symmetric, transitive, and antisymmetric, but it is incomplete whenever X contains two distinct elements.

A.1.2 Important Classes of Relations

Definition A.3. An equivalence relation is reflexive, symmetric, and transitive. A preorder is reflexive and transitive. A partial order is reflexive, transitive, and antisymmetric. A total preorder is a complete preorder. A linear order (or total order) is a complete partial order. A set equipped with a partial order is a partially ordered set, or poset.

In consumer theory, a rational weak preference relation is modeled as a total preorder. Antisymmetry is generally inappropriate for weak preference because distinct alternatives may be indifferent.

Example A.4. The usual relation $\geq$ is a linear order on $\mathbb{R}$. The componentwise relation on $\mathbb{R}^L$, defined by $xRy \iff x_\ell \geq y_\ell$ for every ℓ , is a partial order but is not complete when $L \geq 2$. For example, $(1, 2)$ and $(2, 1)$ are incomparable.

The lexicographic relation on $\mathbb{R}^2$ is defined by $xR_{\text{lex}}y$ if $x_1 > y_1$ or if $x_1 = y_1$ and $x_2 \geq y_2$. It is a linear order. Although complete and transitive, it has no continuous real-valued representation on $\mathbb{R}^2$ with its usual topology.

A.2 Symmetric and Asymmetric Parts

Definition A.5. The symmetric part I_R and asymmetric part P_R of a relation R are defined by

$$xI_Ry \iff xRy \text{ and } yRx, \quad xP_Ry \iff xRy \text{ and } \neg(yRx).$$

Proposition A.6. If R is a preorder, then I_R is an equivalence relation and P_R is asymmetric and transitive.

Proof. Reflexivity and symmetry of I_R follow immediately, and its transitivity follows from transitivity of R . Asymmetry of P_R is built into its definition. If xP_Ry and yP_Rz , then transitivity gives xRz . If zRx , then zRy , contradicting yP_Rz . Thus xP_Rz . $\square$

A.3 Equivalence Classes and Quotient Orders

Definition A.7. Let E be an equivalence relation. The equivalence class of x is $[x]_E = \{y \in X : yEx\}$. The quotient set is $X/E = \{[x]_E : x \in X\}$.

Example A.8. If $X = \{a, b, c\}$ and the only distinct equivalent elements are a and b , then $X/E = \{\{a, b\}, \{c\}\}$. For a preference preorder, the equivalence classes of I_R are indifference classes.

Proposition A.9. Let R be a preorder and let I_R be its symmetric part. On X/I_R , define

$$[x]_{I_R} \bar{R} [y]_{I_R} \iff xRy.$$

Then $\bar{R}$ is well defined and is a partial order.

Proof. Suppose $xI_R x'$, $yI_R y'$, and xRy . Then $x'Rx$, xRy , and yRy' ; transitivity therefore gives $x'Ry'$, which proves that the definition is independent of representatives. Reflexivity and transitivity are inherited from R . If both $[x]\bar{R}[y]$ and $[y]\bar{R}[x]$, then $xI_R y$, so $[x] = [y]$. Thus $\bar{R}$ is antisymmetric. $\square$

A.4 Maximal and Greatest Elements

Definition A.10. For a nonempty subset $S \subseteq X$, the set of **maximal elements** is

$$\text{Max}(S, R) = \{x \in S : \nexists y \in S \text{ such that } yP_R x\}.$$

The set of **greatest elements** is

$$\text{Great}(S, R) = \{x \in S : xRy \text{ for every } y \in S\}.$$

A greatest element dominates every feasible alternative; a maximal element is merely undominated. Every greatest element is maximal. If R is complete on S , then the two notions coincide. Under a partial order there is at most one greatest element, while there may be many maximal elements. Under a preorder, several indifferent alternatives may all be greatest.

Example A.11. Let $S = \{a, b, c\}$ and define $R = \Delta_S \cup \{(a, b)\}$. Thus $aP_R b$ and every other pair of distinct alternatives is incomparable. Then a and c are maximal, but there is no greatest element because a and c are incomparable.

Proposition A.12. *Every nonempty finite subset of a set equipped with a total preorder has at least one greatest element and therefore at least one maximal element.*

Proof. Proceed by induction on the cardinality of S . The claim is immediate for a singleton. Suppose g is greatest in a finite set S , and add $a \notin S$. Completeness gives either aRg or gRa . In the first case, transitivity and gRy for every $y \in S$ imply that a is greatest in $S \cup \{a\}$. In the second case, g remains greatest. $\square$

Finiteness matters. For example, $(\mathbb{R}, \geq)$ is linearly ordered but has no greatest element.

A.5 Transitive Closure

Definition A.13. Let $R^1 = R$, and recursively define $R^{n+1} = R^n \circ R$. The transitive closure of R is

$$R^+ = \bigcup_{n \geq 1} R^n.$$

The reflexive-transitive closure is $R^* = \Delta_X \cup R^+$, where $\Delta_X = \{(x, x) : x \in X\}$.

Proposition A.14. *The relation R^+ is the smallest transitive relation containing R .*

Proof. Concatenating an R -path of length m with one of length n yields a path of length $m + n$, so R^+ is transitive. Any transitive relation containing R must contain every R^n and hence their union. $\square$

A.6 Order Extensions

Definition A.15. Let R and R' be preorders on X . We call R' an extension of R if $R \subseteq R'$ and $P_R \subseteq P_{R'}$. Thus every original weak and strict comparison is preserved.

Theorem A.16. (*Szpilrajn extension theorem*) *Every partial order on a set has a linear-order extension. See [Szpilrajn, 1930].*

Proof. Proof sketch. The standard proof applies Zorn's lemma to the set of partial orders extending the given order, ordered by inclusion; this use of Zorn invokes the axiom of choice. The union of every chain is again a partial order, so a maximal extension exists. The key extension lemma is that if x and y are incomparable, adjoining xRy and taking the transitive closure preserves antisymmetry: a new nontrivial cycle would contain the added edge and would imply an original path from y back to x , contradicting incomparability. Thus two incomparable elements

could be consistently compared, contradicting maximality. Hence the maximal extension is complete and therefore linear. $\square$

Corollary A.17. *Every preorder has a total-preorder extension.*

Proof. Let I_R be the symmetric part of the preorder R . The induced relation $\bar{R}$ on X/I_R is a partial order by the earlier proposition. Next, let $\bar{R}^{\text{ext}}$ be a linear-order extension of $\bar{R}$, and define on X

$$xR^{\text{ext}}y \iff [x]_{I_R} \bar{R}^{\text{ext}} [y]_{I_R}.$$

Then R^{ext} is a total preorder. It contains R , and antisymmetry of $\bar{R}^{\text{ext}}$ ensures that every comparison in P_R remains strict. $\square$

A.7 Checklist for Economic Applications

- Use a total preorder for rational weak preference; use its asymmetric part for strict preference and its symmetric part for indifference.
- Do not call a strict relation a partial order: strict relations are irreflexive, whereas partial orders are reflexive.
- Distinguish maximal from greatest alternatives whenever the relation may be incomplete.
- Use quotient notation X/I_R , not set-difference notation, when collapsing indifference classes.
- State finiteness, compactness, continuity, or another attainment condition whenever existence of a maximal or greatest alternative is claimed.

A.8 Problems

Problems are labelled Core, Proof, or Applied. Core problems consolidate essential techniques; Proof problems develop formal arguments; Applied problems connect the theory to measurement or policy.

Problem A.1 [Core]. Classifying relations.

Assume $|X| \geq 2$. For each relation below, determine whether it is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, transitive, and complete: equality on X ; strict set inclusion on 2^X ; weak set inclusion on 2^X ; and the componentwise order on $\mathbb{R}^2$.

Give a counterexample for every failed property.

Problem A.2 [Proof]. Quotienting a preorder.

Let R be a preorder on X and define xIy if and only if xRy and yRx .

- (a) Prove that I is an equivalence relation.
- (b) Define $[x] \bar{R} [y]$ if and only if xRy . Prove that this definition is independent of representatives.
- (c) Prove that $\bar{R}$ is a partial order on X/I .

Problem A.3 [Proof]. Maximal, greatest, and finite existence.

Let R be a preorder on a nonempty finite set S .

- (a) Prove that S has an R -maximal element.
- (b) Give an example with several maximal elements and no greatest element.
- (c) Prove that if R is complete, maximal and greatest elements coincide.

Problem A.4 [Proof]. Order extensions and multiple representation.

Let R be a preorder on a finite set X .

- (a) Pass to the quotient partial order on X/I_R and extend it to a linear order.
- (b) Pull the extension back to a complete preorder on X .
- (c) Show that R is the intersection of its complete preorder extensions and deduce a finite multiple-utility representation.

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